

Fabrication and characterization of electron microscope 3D resolution standards

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Abstract

Scanning Electron Microscopy (SEM) plays a critical role in nanoscale imaging and analysis, supporting applications in materials science, microelectronics, and nanofabrication. A fundamental parameter that determines the utility of SEM is its spatial resolution, which is the ability to distinguish fine structural details. To evaluate and calibrate this performance effectively, resolution standard samples with well-defined and reproducible features are essential.

This project focuses on the development of a more precise grid pattern resolution standard sample for SEM resolution evaluation. The design takes into steep-edged metallic structures, which serve as reliable references for measuring image sharpness and beam focus. To achieve this goal, lithography technology is used to define fine patterns, followed by controlled metal deposition. Special attention should be paid to minimizing edge related artifacts that may affect resolution measurements.

By analyzing images of the fabricated structures, the resolution of the microscope can be quantitatively assessed through intensity profiles and edge response metrics such as full width at half maximum. The resulting standard sample demonstrates high reproducibility, clear edge contrast, and strong compatibility with high-resolution SEM systems.

Overall, the project has produced a precision structured SEM resolution standard sample, contributing to the evaluation and calibration of electron microscope performance and standardized data.

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1 Introduction

1.1 Background Introduction

1.1.1 Scanning Electron Microscopy

Scanning Electron Microscopy (SEM) is a powerful imaging technique widely employed across disciplines such as materials science, nanotechnology, and semiconductor engineering. SEM uses a finely focused high-energy electron beam to scan the surface of the sample and generate various signal types, the most obvious of which are secondary electrons (SE) and backscattered electrons (BSE), providing information about the morphology and composition of the sample. With magnification capabilities exceeding 300,000X and even up to 1,000,000X in advanced models, SEM enables imaging of features from micrometers down to the nanometer scale with high clarity and depth of field^[1].

Due to its large depth of field and superior spatial resolution compared to optical microscopy, SEM has become indispensable in a wide range of scientific and industrial applications. These include material failure analysis, nanostructure characterization, thin-film inspection, and critical dimension (CD) metrology in semiconductor manufacturing.

1.1.2 Resolution of SEM

Resolution is a key performance metric, which often defines the quality of a scanning electron microscope^[2]. Traditionally, the resolution of a scanning electron microscope is defined as the subjective measurement of the minimum distance between two points on resolution specimen that can still be distinguished as separate entities in the resulting image. It essentially reflects the instrument's ability to resolve fine structural details.

Understanding and quantifying this limit is essential for the accurate analysis of nanoscale materials, especially in applications such as defect inspection, thin-film metrology, and device failure analysis. This capability is especially important in fields such as materials science, nanotechnology and semiconductor device characterization.

1.1.3 Resolution Standard

Resolution standard samples are important tools for evaluating and calibrating instrument spatial resolution in SEM. These samples provide a means to assess the performance of an SEM by offering features of known size, allowing for the evaluation of the system's resolving power. Resolution standards typically contain micro- or nanoscale structures with precisely known dimensions and spacings, such as line arrays or dot patterns. By imaging these standards, we can assess the microscope's ability to resolve fine details under various operating conditions^[3].

By imaging such standards under various accelerating voltages, working distances, and focusing conditions, we can determine the practical resolution limits of the instrument, assess beam spot size, and verify the stability of the electron optical system. Standard samples are also widely used to compare imaging performance across different instruments or experimental sessions, ensuring consistency and reproducibility of data.

In applications involving nanostructure analysis, thin film defect inspection, or semiconductor process evaluation, resolution standard samples play a critical role in validating imaging accuracy and ensuring data reliability.

1.2 Project Plan

In this project, I designed a grid-patterned resolution standard sample, where the blue regions

(as shown in the figure 1.2) represent the deposited metal structures. The metal grid structures were deposited on a substrate in such a way that their well-defined edges could be used to evaluate the resolving power of the SEM. The scanning process involved acquiring high-magnification images across metal–nonmetal interfaces.

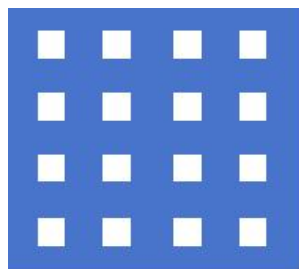


Figure 1.2 Design Structure of Resolution Standard Samples

One effective method to quantify resolution from the SEM image is by analyzing the intensity profile across a sharp metal edge. By fitting the edge profile with a mathematical model, the full width at half maximum(FWHM) of the derivative curve can be extracted. The FWHM can be used to describe the resolution.

1.3 Design Significance

The primary goal of this project is to fabricate nanometer-scale SEM resolution standard samples that can be used to calibrate and assess high-resolution imaging systems. A key design improvement introduced in this project involves minimizing measurement errors caused by non-ideal sample geometry, specifically the sloped sidewalls of metal grid features. In high-resolution SEM analysis, such geometric deviations can lead to inaccurate estimation of beam size or resolution. Therefore, a major focus of this work is on fabricating samples with more vertical metal edges, as flatter and sharper sidewalls reduce artifacts in edge intensity profiles and enhance the reliability of FWHM-based resolution estimation.

1.4 Experimental Scheme

To investigate the influence of fabrication parameters on the edge quality and resolution performance, four grid-patterned samples were prepared, each with varying materials, metal thicknesses, and process conditions. Different lithographic and deposition approaches were employed to assess their impact on edge sharpness and feature definition. By systematically analyzing the structure quality and SEM image results from each sample, the most optimal one was selected for quantitative resolution characterization. Through this comparative analysis, the project aims to determine the ideal fabrication strategy for producing reliable, high-fidelity SEM resolution standards.

2 Lithography

2.1 Basic principles and Methods of Lithography

Lithography encompasses a range of techniques used to transfer patterns with micrometer or nanometer precision onto substrates coated with resist. The basic principle of lithography is to selectively change the chemical solubility of the resist through controlled radiation exposure. When light or particle beams interact with the resist, they induce photochemical or radiolytic reactions changing solubility of resist in the developer solution. This difference in solubility allows for selective removal of exposed or unexposed areas, enabling precise transfer of patterns onto the substrate. These techniques differ primarily in the type of radiation or particle beam used for exposure, as well as their resolution, complexity and suitability for specific applications. For example, lithography methods include ultraviolet lithography, direct laser writing, X-ray lithography, electron beam lithography, ion beam lithography and so on, each with distinct advantages and limitations^[4 5 6].

Lithography stands as one of the most essential techniques in micro- and nanofabrication, with significance that goes far beyond the mere creation of submicron patterns. Due to its high precision, reproducibility, and scalability, lithography provides a reliable foundation for the mass production of complex microstructures. Its value also extends into scientific research and instrumentation, where it plays a critical role in the development of calibration standards and metrology tools. In this project, for example, we employed lithography to fabricate resolution standard samples for SEM, creating metal structures with precisely defined geometries and sharp edges. These serve as robust references for evaluating SEM resolution and image fidelity. This application demonstrates how lithography functions not only as a fabrication method but also as a key enabler of high-accuracy measurement and instrument validation.

2.2 Clean Room Environment in Resist Processing

Resist processing requires clean room environments to prevent contamination that can cause pattern defects. Controlled parameters include airborne particles, humidity, temperature, and airflow. Even tiny contaminants like dust or skin flakes can result in issues such as scumming or delamination.

The key steps of lithography, such as coating, baking, and development, are typically carried out in ISO 6 cleanrooms, while exposure is performed at ISO 5 or higher standards. HEPA filtration ensures laminar flow and positive pressure to reduce particle intrusion.

Operators wear full cleanroom garments to minimize contamination. Stable humidity and temperature are essential, as resist chemistry is highly sensitive to environmental changes.

Particles smaller than 0.5 μm can cause fatal defects in sub-100 nm features^[1].

Clean rooms are thus essential to maintain process stability, resolution, and device yield in modern lithography.

2.3 Sputtering metal layer

2.3.1 Significance of metal layer

Before spin coating resist on a silicon substrate, a thin metal film can be deposited first. When fabricating metal microstructures through electroplating, the patterned areas must have a conductive base to enable the plating reaction. Since resists such as PMMA are insulating and silicon or silicon dioxide is also non-conductive, a pre-deposited metal layer like gold or chromium provides a conductive seed layer. This enables electroplating to occur selectively in exposed regions after lithographic development, supporting the formation of structures such as nickel or copper features^[7].

In some fabrication scenarios, removing the final patterned metal from the substrate without damage is critical. Direct processing on a bare silicon substrate leads to strong adhesion, making release difficult. Introducing a thin metal film as an intermediate layer creates a chemically responsive interface. In the final step, selective etchants can dissolve this metal layer, allowing the clean separation of the patterned structure from the substrate. This approach is particularly useful when fabricating freestanding nanostructures or devices intended for transfer and integration.

2.3.2 Basic principles of magnetron sputtering system

Magnetron sputtering is a widely used physical vapor deposition technique for depositing thin films of metals, semiconductors, and insulators onto various substrates. The system, as illustrated in the figure, includes a vacuum chamber, plasma generation unit, target holder,

and high-voltage power supply. Metal deposition occurs through momentum transfer from energetic argon ions to atoms in the solid target, causing them to be ejected and deposited onto the substrate.

The process starts by evacuating the chamber to a base pressure below 2×10^{-6} Torr using rotary and turbo pumps. Argon gas is introduced, and a high voltage is applied to the target, generating a negative bias that accelerates electrons. These electrons collide with argon atoms, producing Ar^+ ions and initiating plasma formation. A magnetic field generated by a magnetron below the target traps electrons near the target surface, enhancing plasma density and increasing sputtering efficiency.

Ar^+ ions are accelerated toward the negatively charged target and dislodge atoms from its surface. These atoms travel through the low-pressure environment and condense onto the substrate to form a uniform thin film. The basic process is shown in Figure 2.3.2. This technique provides precise control over film thickness, good conformity on complex surfaces, and compatibility with various materials such as Ti, Pt, Ni, and Al. It is also suitable for low-temperature substrates.

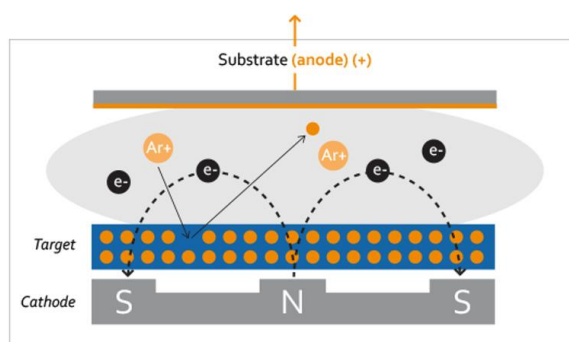


Figure 2.3.2 Basic theory of Magnetron Sputtering

2.3.3 Basic process of magnetron sputtering system

During this experiment, magnetron sputtering was used to deposit metal thin films under a

standardized protocol to ensure safety and reproducibility. Before operation, target materials such as gold, chromium, and copper were selected, and electrical connections were verified. The chamber was vented with nitrogen gas, the system powered off, and the samples mounted onto the stage before resealing the chamber.

Vacuum was initiated by engaging the roughing pump until the pressure dropped below 4.9×10^{-2} Torr. The turbomolecular pump was then activated to achieve a base pressure of 1.9×10^{-6} Torr or lower. Argon gas was introduced at a controlled flow rate of 8 mL/min, and the chamber pressure was stabilized between 2×10^{-3} and 3×10^{-3} Torr. After confirming all shutter positions and cable connections, the MDX sputtering unit was powered on, and deposition was carried out at 0.14 kW. The sample holder rotated during sputtering, and for depositions longer than 100 seconds, cooling intervals were applied.

Upon completion, argon flow and power were shut down, and the chamber was vented again to retrieve the samples. After unmounting, the chamber was resealed and vacuum re-established.

In the formal experiment, a magnetron sputtering system was used to deposit chromium and gold thin films successively on clean silicon substrates of samples 1 and 3. Gold can be used to provide conductivity, while chromium can enhance the adhesion between gold and silicon substrates. By adjusting the sputtering duration, the film thickness can be precisely controlled. The calibrated deposition rate based on the system is approximately 0.3 nm per second. During operation, chromium is sputtered continuously for 10 seconds, followed by gold sputtering for 30 seconds. This is equivalent to a chromium film thickness of about 3nm and a gold film thickness of 9nm. Figure 2.3.3 shows the sample image after magnetron sputtering

of metal.

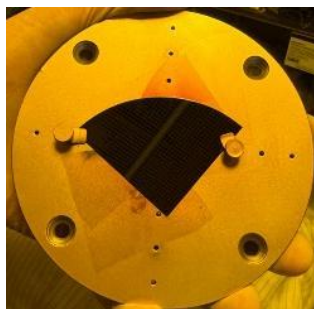


Figure 2.3.3 Sample image after magnetron sputtering of metal

2.4 Spin coating

Spin coating is a widely used technique for producing thin, uniform films on flat substrates, essential in lithography. The process involves depositing a small amount of resist solution onto the substrate, followed by high-speed rotation to spread the liquid and form a thin film. Film thickness and uniformity depend on parameters such as spin speed, resist viscosity, spin time, and ramp profile. Higher spin speeds and lower viscosities result in thinner films, while longer spin durations enhance uniformity due to solvent evaporation.

The process consists of two main phases: viscous flow-driven spreading and evaporation-dominated thinning. Ramp acceleration, ambient conditions like humidity and airflow, and substrate cleanliness all significantly affect the coating quality. A cleanroom environment is typically required to minimize surface defects and ensure consistency.

Proper substrate preparation ensures good wetting and reduces defects such as pinholes and edge irregularities. When optimized, spin coating can produce films ranging from microns to tens of nanometers in thickness, with excellent lateral uniformity. This makes it critical for semiconductor applications where precise film control directly impacts lithographic accuracy and device performance^[1].

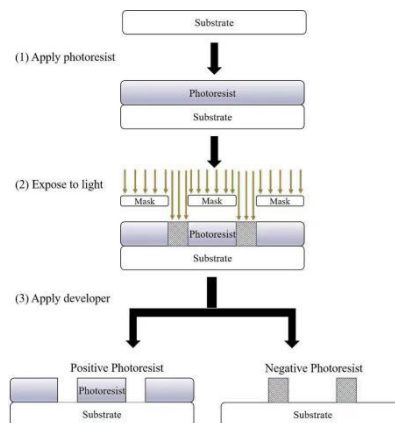
2.4.1 Resist

Resists are essential light- or beam-sensitive materials in micro- and nanofabrication, used to define patterns on substrates. They are classified as positive or negative resists based on their solubility changes upon exposure^[8].

Positive resist is more soluble in the developer after exposure, as light can cause chain breakage and reduce the molecular weight of the polymer^[9]. This causes the exposed regions to dissolve, reproducing the transparent areas of the photomask. Positive resists such as PMMA are widely used in high-resolution applications like electron beam lithography and proton beam writing, due to their sharp edge definition and clean development.

Negative resists behave oppositely. Exposure causes cross-linking, making the exposed regions insoluble. As a result, the unexposed areas are removed, producing an inverse pattern of the mask. Negative resists are favored for thicker structures and applications requiring higher exposure sensitivity, such as MEMS and X-ray lithography, although they typically offer lower resolution and sloped sidewalls.

Figure 2.4.1(a) illustrates the contrast between these two types. Their selection depends on application needs. Positive resists provide finer resolution but demand stricter process control, while negative resists offer mechanical robustness but compromise on pattern fidelity.



2.4.1(a) Different effects of two types of resist

In this experiment, PMMA 950 A11 was selected as the resist for pattern definition. The

molecular formula of PMMA (polymethyl methacrylate) is $(C_5H_8O_2)_n$, and its structure is shown in Figure 2.4.1(b). It is a widely used positive tone resist that reacts to high-energy radiation such as MeV protons or electron beams. Upon exposure, PMMA undergoes chain scission, leading to a reduction in molecular weight and an increase in solubility in the developer. The “950 A11” designation refers to PMMA with an average molecular weight of approximately 950,000 g/mol dissolved in anisole at 11% by weight. This formulation produces a relatively thick resist layer when spin-coated, making it suitable for high-aspect-ratio nanostructure fabrication. The solid form of PMMA has a density of about 1.19 g/cm^3 , and its physical and chemical properties are well-characterized, making it a robust and predictable material for micro- and nanofabrication processes.

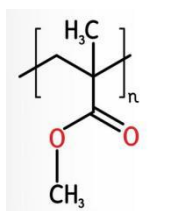


Figure 2.4.1(b) Chemical structure of PMMA

PMMA 950 A11 was chosen for its excellent resolution capability, chemical stability, and compatibility with direct-write techniques such as proton beam writing. It provides a smooth, uniform resist profile with minimal edge roughness, which is critical for accurate pattern transfer. It adheres well to common substrate materials such as silicon, ensuring mechanical stability throughout processing steps including baking, metal deposition, and lift-off. Additionally, its chemical structure is tolerant of various solvents and process environments, which contributes to its effectiveness in multi-step fabrication workflows. The predictable lift-off properties of PMMA also make it a practical choice for applications involving patterned metal structures.

2.4.2 Basic Process of Spin coating

The operation begins by ensuring the spin coater is connected to a stable power supply. If the equipment requires a voltage conversion, a step-up/down transformer is used to adjust the supply voltage appropriately, such as converting 110V to 220V. Once the transformer is switched on, the spin coater can be powered up.

For systems that utilize nitrogen to aid solvent evaporation or maintain a clean environment, the corresponding gas lines must be opened by adjusting the wall-mounted gas valves, and pressure levels should be checked using the attached gauges. Figure 2.4.2 is a physical image of the operating instruments of the spin coating machine in the laboratory. With the gas system ready, the substrate is carefully positioned at the center of the vacuum chuck inside the spin coater. Vacuum is then applied to secure the substrate in place, ensuring it will not move during rotation.

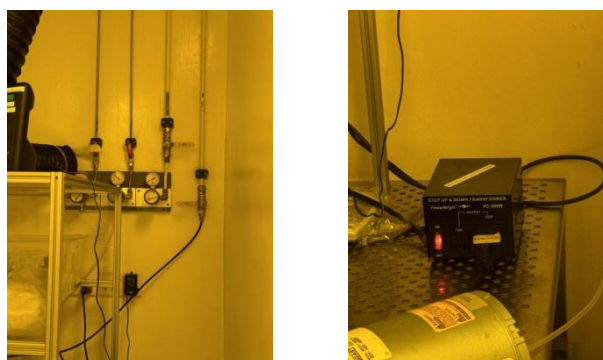


Figure 2.4.2 Physical images of spin coating machine in the laboratory

A measured amount of resist is dispensed directly onto the center of the substrate using a micropipette or dropper. The spin coater parameters, including spin speed, acceleration, and spin time, are then input via the control panel. These parameters are crucial as they directly influence the thickness and uniformity of the resist film. After closing the lid to prevent solvent splash or contamination, the spin coating process is initiated. The resist spreads

outward due to centrifugal force, forming a thin, even coating over the substrate surface. Once spinning is complete, the lid is lifted and the wafer is carefully removed using tweezers.

Finally, the system shuts down in reverse order: turn off the vacuum, close the air valve, and disconnect the power supply. The vacuum chuck is cleaned if any resist residue remains to prepare for the next use. This procedure ensures reproducible and uniform film deposition, which is critical for subsequent lithography steps.

2.4.2 Spin Curve

In this project, PMMA 950 A11 resist was spin-coated onto silicon substrates to fabricate resolution standard samples. The sample reagent is shown in Figure 2.4.2(a). Prior to the main fabrication, a series of spin speed calibration tests were conducted to determine the relationship between spin speed and film thickness for the specific PMMA solution used. Silicon wafers were cleaned and prepared under standard conditions, and PMMA was applied at four different spin speeds: 1000 rpm, 2000 rpm, 3000 rpm, and 4000 rpm.



Figure 2.4.2(a) Sample reagent diagram of PMMA

After each coating step, the thickness of the resulting PMMA film was measured using cross-sectional SEM imaging. The obtained values were plotted to generate a spin curve, as shown in the figure 2.4.2(b) below. This curve showed the expected inverse square root dependence of film thickness on spin speed. This curve served as an essential reference for selecting the optimal spin parameters for the final fabrication process.

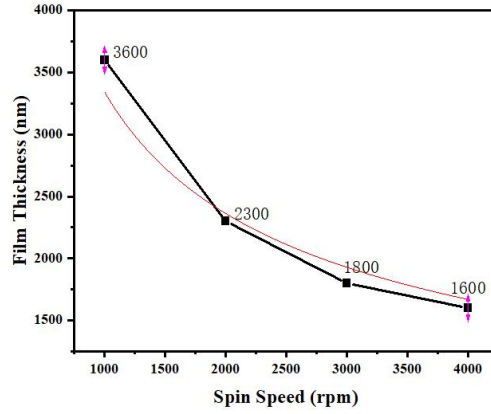
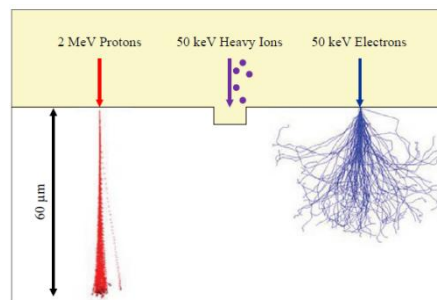


Figure 2.4.2(b) Spin curve of PMMA resist

2.4.3 Simulation

1. Confirmation of lithography method

Before starting the spin coating process, as shown in the figure 2.4.3(a), the behaviors of three different beams—2 MeV protons, 50 keV heavy ions, and 50 keV electrons—are compared as they pass through a 60 μm thick resist layer. Among them, the electron beam exhibits significant lateral scattering, causing the exposed area to spread and lose pattern precision. Heavy ions scatter less than electrons but deposit a large amount of energy in a small volume, which may damage the resist or even the substrate. In contrast, protons travel along nearly straight paths with minimal lateral scattering, resulting in highly localized energy deposition and well-defined pattern edges.



2.4.3(a) Comparison between proton beam, focused ion beam, and electron beam writing^[10]

Then, we conducted beam scattering simulations using two widely recognized Monte Carlo tools: CASINO for electron beams and SRIM for proton beams. The goal is to determine the

optimal lithography method to minimize lateral scattering and generate vertical sidewalls, which was an essential factor for accurate resolution measurements of the fabricated grid structures. Maintaining steep and well-defined sidewalls is critical in SEM calibration structures, especially when evaluating the resolution limit of scanning techniques. Any significant sidewall tilt would introduce uncertainty in feature edge detection, leading to measurement errors.

The CASINO simulation results, as shown in the figure 2.4.3(b), demonstrate that 100 keV electron beams undergo substantial lateral spreading as they propagate through PMMA resist layers, regardless of the thickness. As the PMMA thickness increases, this effect becomes even more pronounced, ultimately preventing the formation of straight and vertical features in the resist.

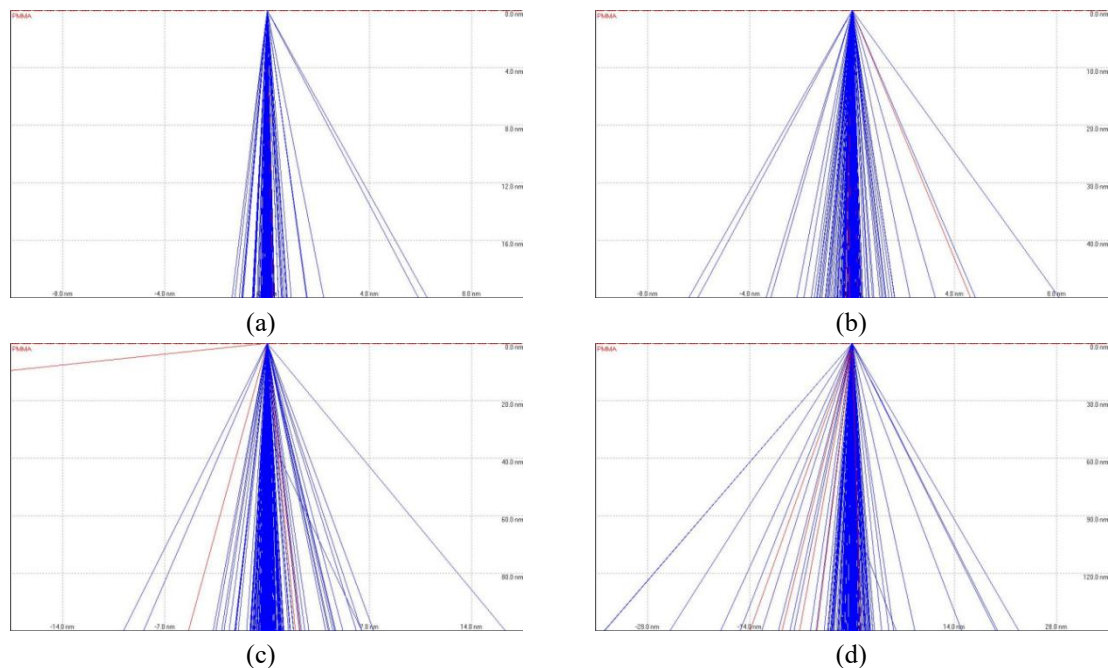


Figure 2.4.3(b) The straggling distance of 100keV electron beam in PMMA with thicknesses of (a) 20nm, (b) 50nm, (c) 100nm and (d) 150nm

The SRIM simulation results provide a clear comparison of proton beam behavior in PMMA resists of different thicknesses under 2000keV irradiation. As shown in the figure 2.4.3(c), in

the 10 μm thick PMMA sample, the proton trajectories remain nearly collimated along the beam axis, showing negligible lateral deviation. The exposure profile is highly confined, with energy deposition concentrated in a narrow vertical track, which is ideal for forming steep and well-defined sidewalls. When the PMMA thickness increases to 100 μm , the proton beam still maintains excellent directional stability. Although the penetration depth becomes significantly greater, the beam continues to exhibit minimal lateral scattering. This confirms that even over long travel distances within the resist, protons retain their linear trajectory, enabling deep and precise patterning.

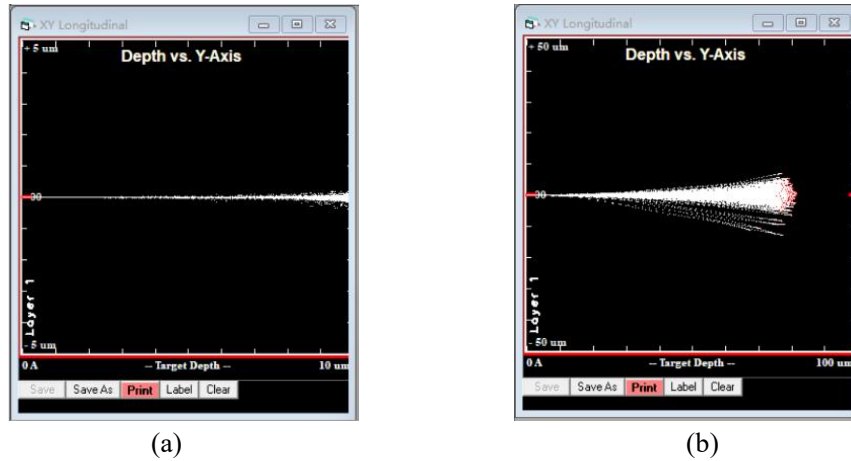


Figure 2.4.3(c) The straggling distance of 2000keV proton beam in PMMA with thicknesses of (a) 10 μm and (b) 100 μm

Based on these simulation results, proton beam writing was chosen as the fabrication method for this experiment. Its superior beam confinement ensures vertical sidewalls and reduces pattern distortion, thereby significantly enhancing the fidelity and resolution of the fabricated nanostructures.

2. Confirmation of spin coating thickness

To ensure high-resolution structure fabrication in Proton Beam Writing, achieving vertical sidewalls in the patterned features is essential. Sidewall angles directly influence the fidelity of the pattern and the accuracy of subsequent beam size measurements during scanning. To

this end, we performed a simulation study to evaluate how different resist thicknesses and metal layer deposition conditions affect the final sidewall slope of the fabricated structures.

To investigate how different metal deposition conditions influence the final sidewall slope, a simplified geometrical model was established. As shown in figure 2.4.3(d), in this model, the incident proton beam travels vertically through the PMMA resist and experiences lateral straggling as it propagates. After development and subsequent metal deposition, the extent of this lateral spread relative to the resist thickness gives rise to a measurable sidewall angle. The relationship is governed by the ratio between the straggling distance and the depth of the PMMA, forming an angle that characterizes the deviation from an ideal vertical edge.

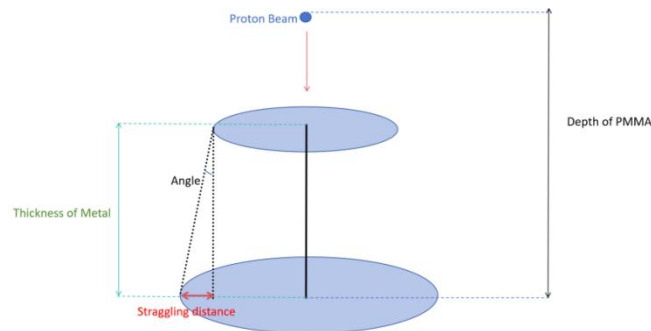
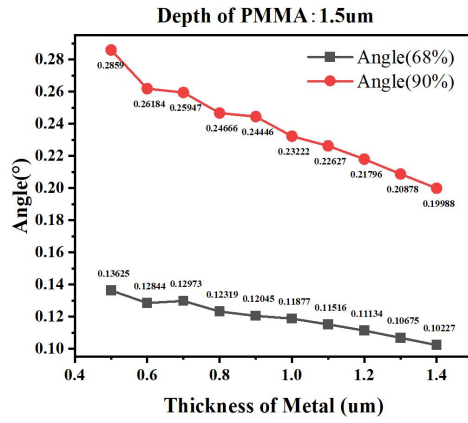


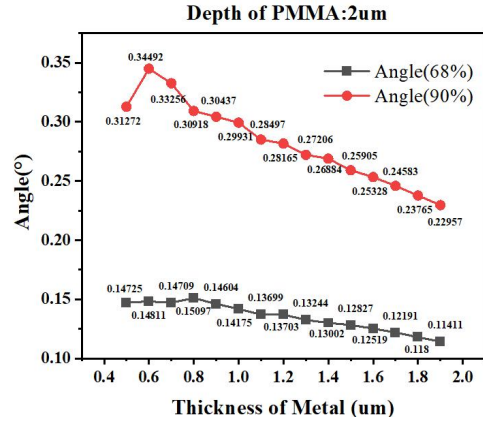
Figure 2.4.3(d) Model for simulating straggling distance and angle

The simulation focused on proton beam interaction with PMMA resist layers of varying thicknesses, ranging from 1.5 μm to 7 μm . For each resist thickness, the lateral straggling of the proton beam and the effective angle of the resulting metal sidewall were calculated after metal deposition. The results are shown in Figure 2.4.3(e), where the corresponding sidewall angles covering 68% and 90% of the proton distribution are plotted respectively. As the PMMA thickness increased, the angular deviation of the deposited metal sidewall became more pronounced. This can be attributed to the broader lateral energy distribution of protons as they travel through thicker resist layers, leading to less confined energy deposition and

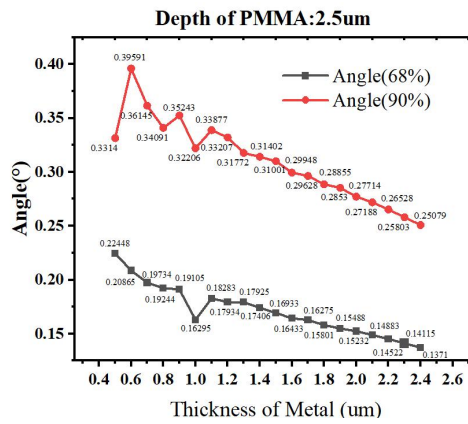
more tapered profiles.



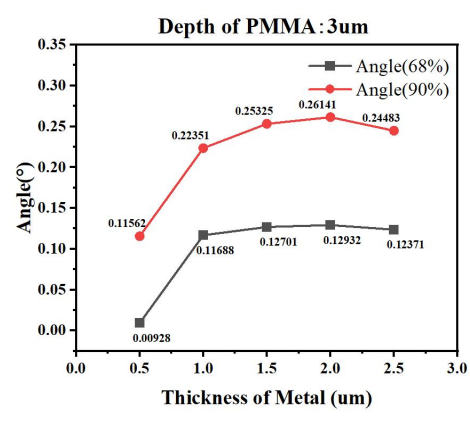
(a)



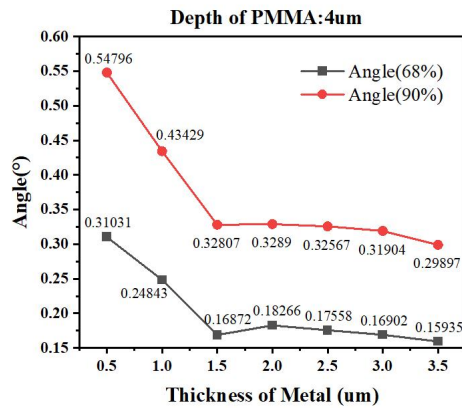
(b)



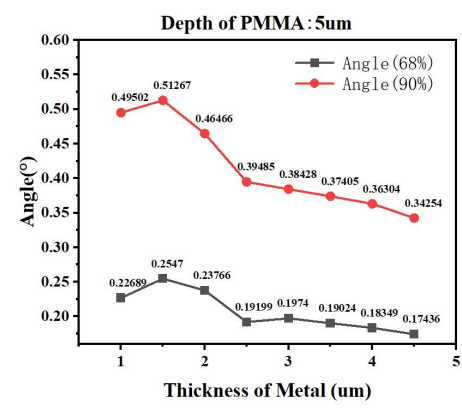
(c)



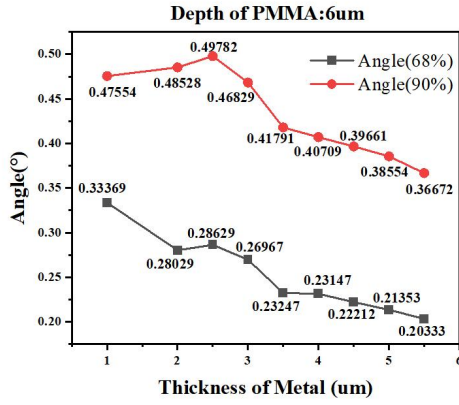
(d)



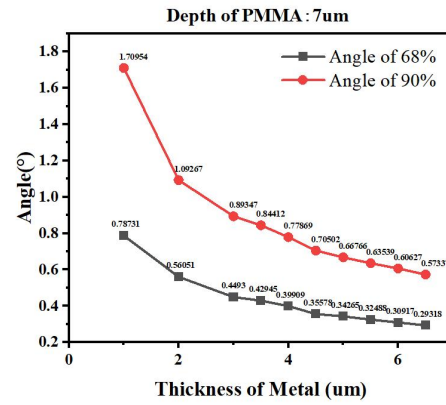
(e)



(f)



(g)



(h)

Figure 2.4.3(e) Deposition of different thicknesses of metal sidewall angles in PMMA with thicknesses of (a)1.5µm,(b)2µm,(c)2.5µm,(d)3µm,(e)4µm,(f)5µm,(g)6µm and (h)7µm

Among the simulated cases, a PMMA thickness of approximately 1.5 µm resulted in the smallest sidewall angle, indicating the most vertical structure. This outcome confirms that thinner PMMA layers are more favorable for preserving structural integrity and producing near-vertical sidewalls after development and metallization.

This simulation-guided approach ensures that the fabricated calibration structures will exhibit minimal edge sloping, which is critical for accurate scanning electron microscopy analysis and beam size extraction in PBW applications.

Therefore, the spin-coating process was carried out using the instrument shown in the figure 2.4.3(f). The spin coater was set to operate at a rotation speed of 4000 rpm for a duration of 1 minute. This procedure ensured uniform spreading of the PMMA resist across the substrate surface, forming a consistent thin film essential for subsequent lithographic steps.



Figure 2.4.3(f) Operation diagram of spin coating process

2.5 Softbake

2.5.1 Basic principles of softbake

Softbake is a critical step in lithography, performed after spin coating to remove residual solvents and harden the resist. This process enhances both chemical stability and mechanical strength of the resist film, improving its adhesion to the substrate and reducing standing wave effects. These outcomes are vital for achieving high-resolution, defect-free patterns.

Typically conducted on a hotplate at 90 °C to 110 °C for one to two minutes, the bake conditions depend on the resist type and thickness. Insufficient baking can leave solvents that lead to poor adhesion or development issues, while excessive baking may induce thermal crosslinking that reduces sensitivity and resolution. Consistent thermal contact is essential to ensure uniform film performance across the wafer. For advanced lithography using chemically amplified resists or multilayer stacks, precise bake control is even more important, as slight deviations may affect feature profile or resolution^[11].

Overall, softbake prepares the resist for accurate pattern transfer and plays a key role in achieving uniformity, resolution, and process yield in modern lithography^[12].

2.5.2 Operation process of softbake

After spin coating the PMMA resist onto the silicon substrate, a soft bake process was conducted to remove residual solvent and enhance film adhesion. After spin coating resist PMMA, the substrate was immediately transferred to a pre-heated hotplate set at 180 °C. The resist-coated wafer was placed flat and centered on the hotplate to ensure uniform thermal contact, and the soft bake was performed for 90 seconds, as shown in figure 2.5.2. This step ensured adequate solvent removal, minimized potential swelling during exposure, and

improved the mechanical stability of the resist. Throughout the baking process, the sample remained undisturbed to avoid introducing thermal gradients or non-uniformities in resist performance. This treatment produced a smooth, dry, and well-adhered resist layer, ready for subsequent proton beam exposure.



Figure 2.5.2 Operation diagram of softbake process

2.6 Exposure

2.6.1 Basic principles of exposure

Exposure is a critical step in lithography where the mask pattern is transferred to the resist through controlled radiation. In positive resists, the exposed regions become more soluble due to photochemical chain scission, while in negative resists, cross-linking makes illuminated areas insoluble^[13]. Accurate pattern transfer depends on delivering a uniform energy dose, determined by light intensity and exposure time. Underdosing leads to incomplete development, and overdosing may reduce resolution or distort profiles.

Maintaining uniform irradiation and system stability is essential. The wafer must remain flat to avoid vibration-induced distortion, and optical systems must reduce light scattering and reflections to preserve contrast. Antireflective coatings and collimated beams help maintain uniformity. Alignment is also crucial, especially for multilayer structures. Misalignment causes overlay errors that compromise device performance. Advanced exposure systems employ automatic alignment to achieve nanometer-level precision. Alternative exposure

sources like electron or ion beams offer higher resolution due to their shorter wavelengths and non-diffraction-limited optics^[14].

Ultimately, exposure defines the fidelity of the latent pattern and sets the resolution limit for all subsequent processes. Precise dose control, system stability, and accurate alignment ensure successful pattern transfer and form a foundation for high-yield micro- and nanofabrication.

2.6.2 Basic introduction of Proton Beam Writing

In this project, Proton Beam Writing was employed as the primary lithographic method to fabricate high-resolution grid structures on silicon substrates. The technique was selected for its excellent beam stability, high pattern fidelity, and direct-write capabilities. PBW utilizes focused MeV protons to induce localized ionization and polymer chain scission within the resist, typically polymethyl methacrylate (PMMA), thereby enabling solubility contrast in exposed regions. Unlike optical or electron beam lithography, PBW is maskless and proximity-effect-free due to the straight trajectories of protons, which arise from their high mass and low lateral scattering.

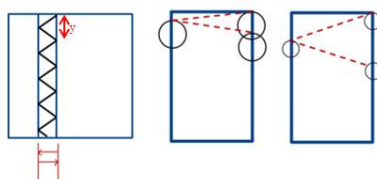
This allows for precise definition of submicron features with near-vertical sidewalls and minimal edge distortion, making PBW especially suitable for applications requiring high-aspect-ratio structures. In-situ imaging based on secondary electron (SE) signals generated during proton impact further enhances beam alignment and resolution control, improving pattern reproducibility and throughput^[15].

Compared to conventional exposure methods in lithography, which rely on light or electron-induced chemical changes in the resist^[13 14], PBW offers superior resolution and structural clarity. These advantages make it a powerful tool for fabricating nanoscale features

in research areas such as photonic devices, microfluidics, and X-ray optics.

2.6.3 Test sample of PBW

Before the final fabrication process, a set of test patterns was written to evaluate the effect of beam parameters on structure quality in PBW. In the PBW process, the proton beam repeatedly scans back and forth across the exposure area, as shown in the figure 2.6.3(a). If the stage moves a smaller distance than the beam spot size during each scan cycle, the beam overlaps with the previously exposed area. This results in a more continuous and smoother edge profile. However, if the stage moves too far in one cycle, the exposed lines won't overlap, and gaps may appear, making the edge rough.



2.6.3(a) Schematic diagram of proton beam lithography process

Based on this principle, I designed several test samples to explore this effect. The table 2.6.3(a) below summarizes the relevant parameters, five key parameters were used to define the exposure conditions. The proton dose, expressed in charge per unit area, determines the amount of energy delivered to the resist and influences how slowly the stage must move to ensure proper exposure. The beam frequency defines the rate at which the proton beam delivers pulses, and it was kept constant throughout the test. The stage speed controls how fast the substrate translates beneath the beam during scanning and was adjusted to match each dose. Additionally, the number of pulses per scan line and the total scanning length of each line were fixed, allowing the calculation of the beam scanning time based on the pulse count and frequency. From these parameters, the beam spot scanning time, the duration needed for

the beam to traverse one scan line, was derived. The proton beam completes a round-trip scan containing 1024 pixels at a scan frequency of 50 kHz. That means each back-and-forth cycle takes about 20 microseconds. Combining this with the known stage speed allowed calculation of the stage movement per dwell period, which represents how far the substrate moves vertically while the beam completes a single horizontal line scan. This value was then compared to the beam spot diameter to determine whether sufficient overlap was achieved.

Table 2.6.3(a) PBW parameter setting of test sample

	dose (nC/mm ²)	Frequency (Hz)	Stage speed(mm/s)	y(μ m)
test 1	150	50000	0.02358	0.4829184
test 2	200	50000	0.01769	0.3622912
test 3	250	50000	0.01415	0.289792
test 4	300	50000	0.01179	0.2414592
test 5	350	50000	0.01011	0.2070528
test 6	400	50000	0.00884	0.1810432

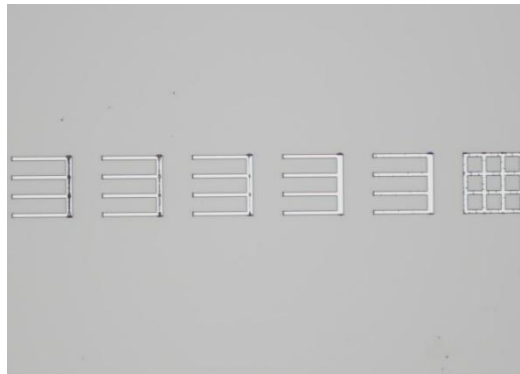


Figure 2.6.3(b) Test samples observed under optical microscope

Comparing this movement distance with the beam spot size, I find that in the last three samples, the movement is smaller than the spot size, the beam spot size setting is shown in the table 2.6.3(b). As a result, the overlapping exposure produced sharp and continuous structures with well-defined edges. Conversely, for lower doses, the stage moved more than the spot size during each dwell period, leading to incomplete or distorted features with poor edge quality.

These differences are visually apparent in the bottom image, which shows test patterns fabricated under varying dose conditions.

Table 2.6.3(b) Beam spot size parameter setting of PBW

	X (um)	Y (um)
Scan size	12.3	12.9
Beam spot size	0.25	0.35

2.6.4 Operation process of PBW

During the actual PBW process, the parameter settings were carefully selected based on preliminary test exposures to ensure smooth edges and high structural fidelity. The specific parameters used are listed in the table: a beam dose of 400 nC/mm², a scanning frequency of 50,000 Hz, and a stage speed of 0.00884 mm/s. These parameters calculated the beam dwell time per spot and stage moving distance of 0.1810432 μ m, which is smaller than the beam spot sizes in both the X (0.35 μ m) and Y (0.4 μ m) directions. This ensured sufficient overlap between adjacent beam spots, enabling continuous and high-quality pattern formation. In addition, the slit and collimator settings were optimized during the exposure process, as shown in table 2.6.4(c): the objective slit width was set to 60 μ m in X direction and 40 μ m in Y direction, while the collimator dimensions were 80 μ m in both directions. These settings resulted in angular spreads of 0.020 in X direction and 0.003 in Y direction, which supported stable beam focusing and high-resolution patterning.

Table 2.6.4(a) PBW parameter setting of sample

dose (nC/mm ²)	Frequency (Hz)	Stage speed(mm/s)	y(μ m)
400	50000	0.00884	0.1810432

Table 2.6.4(b) Beam spot size parameter setting of PBW

	X (um)	Y (um)
Beam spot size	0.35	0.4

Table 2.6.4(c) The setting of objective slit and collimator dimension

Slit setting				Angle spread	
Obj x/um	Obj y/um	Coll x/um	Coll y/um	x	y
60	40	80	80	0.020	0.003

Under this setting, a well-defined square array was successfully fabricated, with each square measuring approximately $255\text{ }\mu\text{m} \times 255\text{ }\mu\text{m}$. The width of the exposure area is $5\text{ }\mu\text{m}$, and the width of the spacing area is $20\text{ }\mu\text{m}$. Figure 2.6.4 illustrates the basic structure of the sample, with the gray area representing the exposure area. The uniformity and clarity of the features demonstrate the effectiveness of the selected parameters and overall process stability.

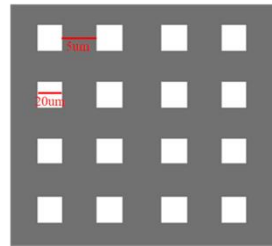


Figure 2.6.4 Schematic diagram of sample structure after PBW

2.7 Development

2.7.1 Basic principles of development

The development step in lithography is critical for converting the latent image formed during exposure into a physical relief pattern on the resist. In positive resists, exposed regions become soluble due to photochemical chain scission, while in negative resists, cross-linking reduces solubility. Development therefore amplifies the contrast created during exposure into well-defined topographical structures.

Development usually involves immersing the chip in a developer solution and selectively

dissolving it based on the exposure state of the resist. Factors such as developer type, concentration, temperature, and duration significantly influence pattern fidelity. Agitation or spray methods are often used to enhance uniformity. Overdevelopment may erode fine features, while underdevelopment can lead to incomplete pattern formation. A post-development rinse in deionized water halts chemical reactions and removes residues, preventing defects such as scumming or bridging. Proper rinsing and drying are especially important for maintaining the integrity of high-aspect-ratio structures^[16].

2.7.2 Operation process of development

In this experiment, the development of the PMMA resist was carried out using a standard developer solution consisting of isopropyl alcohol (IPA) and Distilled water in a 7:3 volume ratio. After exposure, the sample was gently immersed in the developer for 3 minutes at room temperature without agitation. This mixture is known to selectively dissolve the exposed areas of PMMA due to chain scission caused by high-energy proton irradiation, resulting in a clear and high-contrast pattern.

The development time and solvent composition were carefully chosen to ensure complete removal of the exposed regions while minimizing swelling or damage to the unexposed areas. Following development, the sample was rinsed thoroughly in pure IPA to stop the development reaction and remove any remaining residues. This step was crucial to maintaining edge sharpness and dimensional accuracy of the final pattern, which is particularly important for subsequent metal deposition and SEM-based resolution analysis.

The structures of samples after development are shown in the figure 2.7.2.

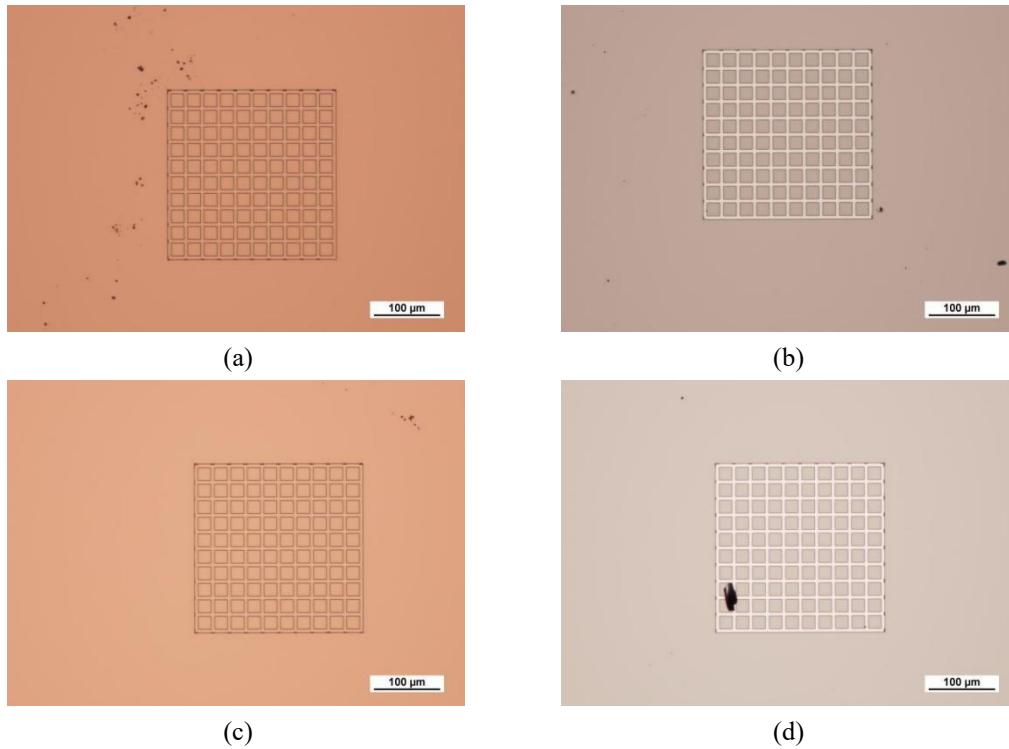


Figure 2.7.2 Grid like structures of (a) sample 1,(b) sample 2,(c) sample 3 and (d) sample 4 under optical microscope

2.8 Metal Plating

2.8.1 Metal selection

In SEM-based nanofabrication processes, especially during the post-development metal plating step, selecting a suitable metal is critical to ensuring high-quality imaging, structural fidelity, and reproducibility of experimental results. This section evaluates six key material properties — oxidation resistance, hardness, proton number, density, cost and magnetic properties — for three commonly used metals: platinum (Pt), gold (Au), and nickel (Ni). The comparative data for these materials are shown in the table 2.8.1 below.

Table 2.8.1 material properties of Pt, Au and Ni

	Au	Pt	Ni
Oxidation-Standard electrode Potential E° (V)	+1.50	+1.20	-0.25
Hardness (HV)	30	40	64
Density (g/cm^3)	19.32	21.45	8.9
Proton Number	79	78	28
Cost	High	Medium	Low
Magnetism	Non-magnetic	Non-magnetic	Strong Ferromagnetism

Oxidation resistance is vital for maintaining surface quality during storage or SEM imaging.

Metals prone to oxidation form insulating oxides that degrade conductivity and introduce imaging artifacts like distortion and poor contrast. Platinum and gold, with standard electrode potentials of +1.20 V and +1.50 V, exhibit excellent oxidation resistance. In contrast, nickel (-0.25 V) is more reactive and less stable under ambient or humid conditions due to its greater tendency to oxidize.

Mechanical hardness influences both coating durability and surface uniformity. Nickel has the highest Vickers hardness (64 HV), followed by platinum (40 HV) and gold (30 HV). Although nickel is harder, it is also more brittle and may produce rough surfaces. Gold's softness increases its vulnerability to scratches, potentially affecting SEM contrast and pattern fidelity.

Atomic number and density play critical roles in SEM imaging contrast. Gold ($Z=79$) and platinum ($Z=78$) generate stronger secondary and backscattered electron signals than nickel ($Z=28$), enhancing image clarity. Additionally, platinum's high density (21.45 g/cm^3) better attenuates electron penetration than gold (19.32 g/cm^3) or nickel (8.9 g/cm^3), reducing substrate interference and improving imaging resolution.

Magnetic behavior strongly affects the stability of the electron beam during SEM imaging. Nickel, being ferromagnetic, can interfere with electron beam paths and cause imaging instability. Platinum and gold are non-magnetic, making them more suitable for stable high-resolution SEM imaging without beam deflection or artifacts.

In terms of cost, nickel is the most economical, but its performance limitations make it less ideal for SEM applications. Gold provides superior performance but at a prohibitive cost for regular use. Platinum offers a balanced solution, combining good oxidation resistance, imaging performance, and manageable cost, making it a practical choice for nanoscale coatings in SEM-related processes.

Considering the factors above, platinum offers a favorable combination of oxidation resistance, mechanical stability, imaging compatibility, and cost-efficiency. While gold excels in conductivity and anti-oxidation, its high cost limits its practicality. Nickel, despite its hardness and affordability, poses risks due to its magnetic properties and susceptibility to oxidation. Therefore, platinum emerges as the optimal choice for post-development metal coating in SEM-based nanostructure fabrication.

2.8.2 Basic principle of Plasma Sputtering

Among the various deposition techniques available, argon plasma sputtering is one of the most commonly employed methods for preparing conductive coatings suitable for SEM analysis. The system used in this work is the JEOL JFC-1600 Auto Fine Coater, as shown in figure 2.8.2(a), a compact and precise sputtering device specifically designed for electron microscopy sample preparation.



Figure 2.8.2(a) Operation diagram of Auto Fine Coater

The principle of sputter coating is based on physical vapor deposition, where atoms are ejected from a solid metal target and deposited onto a substrate. In argon plasma sputtering, high-purity argon is introduced into a vacuum chamber containing a metal target such as platinum or gold. A high DC voltage ionizes the argon gas, and the resulting Ar^+ ions are accelerated toward the negatively charged target. These ions strike the target surface, transferring momentum and dislodging metal atoms. The ejected atoms travel through the vacuum and condense on the substrate, gradually forming a thin, uniform metal film. As shown in Figure 2.8.2(b), the resulting coating is typically fine-grained and amorphous, ensuring good conductivity with minimal topographical interference. The process is entirely physical and allows for precise thickness control by tuning parameters such as coating time, discharge current, and gas pressure.

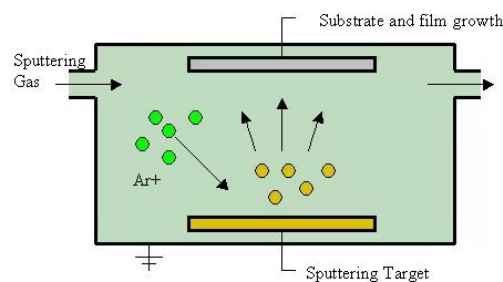


Figure 2.8.2(b) The specific working principle of Plasma Sputtering

The JEOL JFC-1600 sputter coater supports both AUTO and MANUAL modes. AUTO mode automates gas regulation, plasma ignition, and coating duration, making it ideal for routine use. MANUAL mode allows full user control over parameters like discharge time and voltage,

which is useful for custom experiments. Both modes follow the same sputtering principle but differ in automation level.

During coating, the sample is mounted and the chamber is evacuated to 0.03 – 0.06 Pa. Argon is introduced, and parameters such as discharge current and time are set. Plasma discharge produces a visible glow, confirming ignition. Once the preset time completes, the chamber is vented and the coated sample is retrieved.

The instrument includes a calibration curve that links discharge settings to film thickness, enabling precise control. Cleanliness of the sample, holder, and target is critical, as contamination or improper vacuum can affect film uniformity and adhesion. When properly maintained, the JEOL JFC-1600 consistently produces high-quality conductive films suitable for SEM analysis.

2.8.3 Operation process of Plasma Sputtering

In this experiment, platinum (Pt) was deposited on patterned samples to achieve two distinct film thicknesses: 20nm and 2nm. The appropriate sputtering conditions were determined using the JEOL JFC-1600 coating thickness reference chart 2.8.3(a), which correlates film thickness with discharge current and sputtering time.

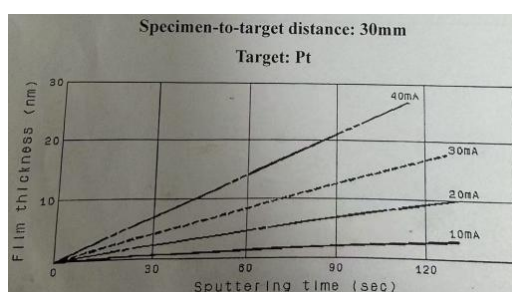


Figure 2.8.3(a) Relationship between sputtering time and film thickness

For the 20 nm Pt coating, a discharge current of 40 mA was selected, with the reference chart indicating a required sputtering time of approximately 85 seconds at a target-to-sample

distance of 30 mm. However, since the substrate contained a PMMA resist layer, which is sensitive to temperature, care had to be taken to avoid thermal damage. Prolonged plasma exposure during sputtering can raise the sample temperature, potentially softening or melting the PMMA and distorting the microstructures. To mitigate this, the sputtering was performed in four intervals with active cooling pauses. Specifically, the system was operated for 20 seconds per cycle, followed by a 1-minute cooling period with chamber venting. The final cycle was extended to 25 seconds, resulting in a total sputtering duration of 85 seconds, as required. The sample structure after platinum plating is shown in the figure 2.8.3(b).

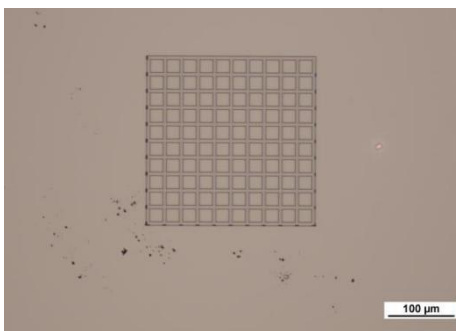


Figure 2.8.3(b) Sample 1 after platinum plating under optical microscope

For the 2 nm Pt coating, which was intended to serve as a minimal conductive layer for surface imaging, a discharge current of 10 mA was used. According to the thickness chart, this required a sputtering duration of approximately 40 seconds. Again, to prevent thermal deformation of the PMMA structures, the process was split into short intervals: the system was operated in four 10-second sputtering cycles, each followed by a 1-minute cooling break. The sample structure after platinum plating is shown in the figure 2.8.3(c). This staged sputtering approach ensured uniform metal deposition while minimizing thermal stress and preserving the integrity of the underlying resist pattern.

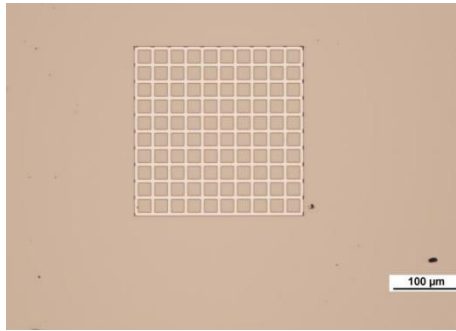


Figure 2.8.3(c) Sample 2 after platinum plating under optical microscope

2.8.4 Basic principle and process of Electron Beam Evaporation

In this experiment, platinum thin films were deposited using an AJA International UHV Electron Beam Evaporation System (Model: EMOC-380), which operates based on the principles of high-vacuum electron beam physical vapor deposition (EB-PVD). This method enables the deposition of high-purity and well-controlled metallic thin films, and is especially suitable for noble metals like Pt due to its high melting point and resistance to oxidation.

The fundamental principle of electron beam evaporation involves the use of a focused, high-energy electron beam to heat and vaporize a solid target material. The electron beam is typically generated by thermionic emission from a tungsten filament and accelerated by a high voltage, then directed using magnetic fields onto the surface of the Pt source held in a water-cooled crucible. Upon impact, the kinetic energy of the electrons is converted into thermal energy, leading to localized heating and rapid evaporation of the Pt atoms into the vacuum chamber^[17].

Under ultra-high vacuum conditions, the evaporated Pt atoms travel in straight-line paths due to minimal gas-phase collisions and condense onto the cooler substrate surface positioned above the crucible. The deposition rate and final film thickness can be precisely controlled by adjusting the beam power and monitoring the deposition using quartz crystal microbalance (QCM) sensors or other in-situ thickness monitors^[18]. The working principle is shown in

Figure 2.8.4.

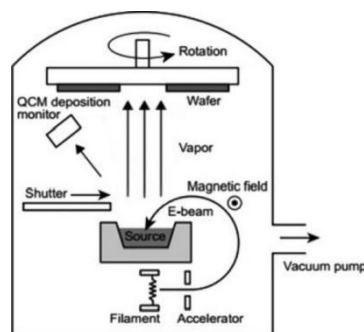


Figure 2.8.4 The working principle of Electron Beam Evaporation

In electron beam evaporation, the vapor flux primarily propagates along straight-line trajectories, enabling highly directional deposition. This allows the formation of smooth interfaces and steep sidewalls, making the technique particularly suitable for metal deposition following micro- and nanolithography processes that require high conformality.

In this project, electron beam evaporation was employed to deposit a 50 nm platinum layer onto the surfaces of Sample 3 and Sample 4. The electron beam evaporation process was performed under a high-vacuum environment, with the chamber evacuated to a base pressure of approximately 7.5×10^{-6} Torr to minimize contamination and ensure clean deposition conditions. Titanium was first deposited as the adhesion layer at a rate of $0.5 \text{ \AA/s}$ to a thickness of 5 nm. Subsequently, platinum was deposited at a slower rate of $0.2 \text{ \AA/s}$ to a final thickness of 50 nm, ensuring improved film uniformity. The entire deposition sequence was conducted without breaking vacuum between the Ti and Pt layers to avoid interfacial contamination. This step was critical for enabling subsequent characterization and improving the imaging contrast in scanning electron microscopy. As shown in Figure 2.8.4, the optical microscope image of Sample 3 after Pt deposition reveals the overall structure and uniformity of the deposited layer.

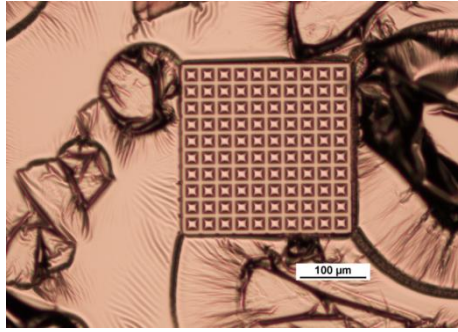


Figure 2.8.4 Sample 3 after platinum plating under optical microscope

2.9 Lift-off

2.9.1 Basic principles of lift-off

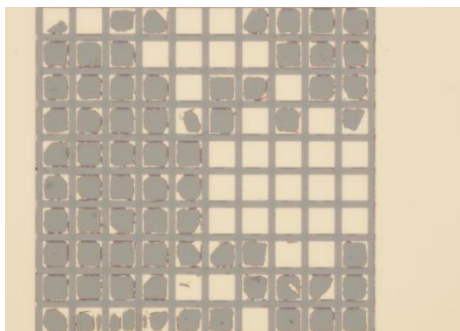
The lift-off process is a standard technique for patterning metal films in micro- and nanofabrication. It begins with resist patterning, followed by metal deposition over the entire surface. A solvent such as acetone or NMP then dissolves the underlying resist, removing the overlying metal and leaving the desired pattern only on exposed substrate areas. Since the metal is not chemically bonded to the resist, it detaches as the resist dissolves. A resist profile with steep or undercut sidewalls is important to prevent metal bridging, which can obstruct clean removal. Gentle heating or ultrasonic agitation may assist the lift-off process, though excessive force can damage delicate structures^[9].

2.9.2 Operation process of lift-off

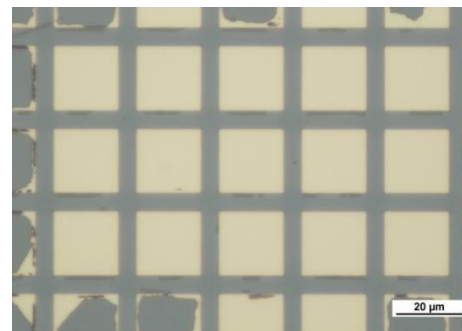
In this experiment, the removal of the residual PMMA resist after metal deposition was achieved using a lift-off technique. The developer used was acetone, a common organic solvent with strong solvency for PMMA. The sample was first immersed in acetone and allowed to soak to initiate the dissolution of the exposed PMMA layer. This step facilitates the lift-off of the metal deposited on top of the resist. Following the initial soak, the sample was subjected to additional immersion in acetone with ultrasonic agitation. The use of ultrasonic energy helps accelerate the dissolution process by mechanically agitating the solution,

promoting the detachment of any remaining metal and resist fragments.

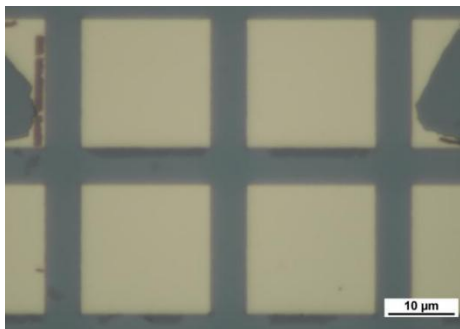
The optical microscope images 2.9.2 below show the surface condition of each sample after the lift-off process. Sample 1 was soaked in acetone for 4 minutes and then subjected to ultrasonic agitation for 2 hours. Sample 2, in contrast, underwent direct ultrasonic agitation for 30 minutes without a pre-soaking step. Samples 3 and 4 followed a consistent protocol involving 5 minutes of soaking followed by 2.5 hours of ultrasonic treatment.



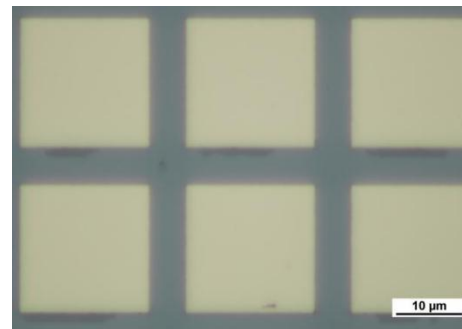
(a)



(b)



(c)



(d)

2.9.2(a) (a) 20 times, (b) 50 times and (c)(d) 100 times of sample 1 after lift-off under optical microscope



(a)



(b)

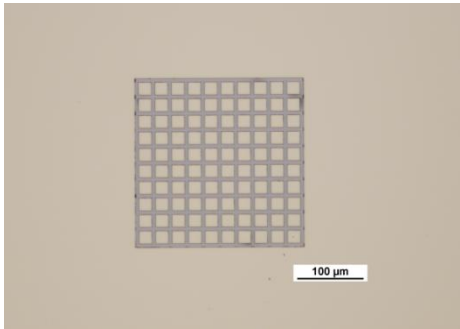


(c)

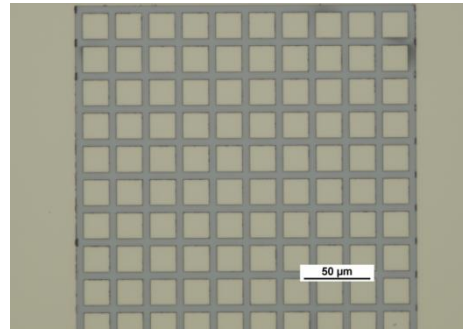


(d)

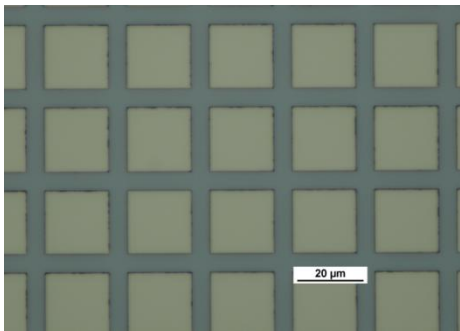
2.9.2(b) (a) 5 times, (b)10 times, (c) 20 times and (d) 50 times of sample 2 after lift-off under optical microscope



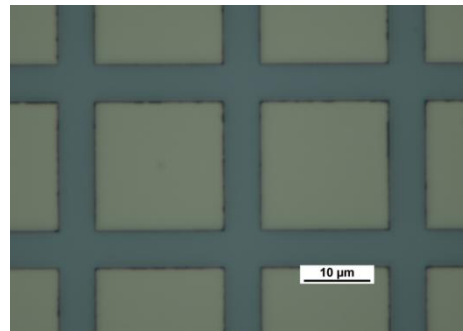
(a)



(b)

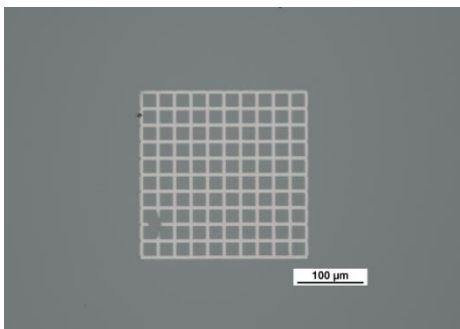


(c)

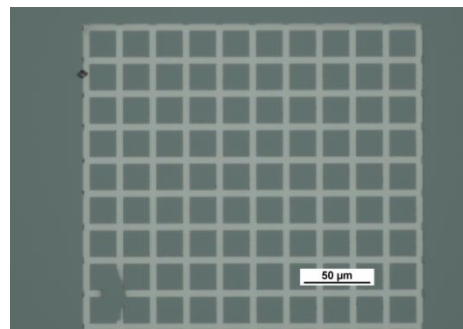


(d)

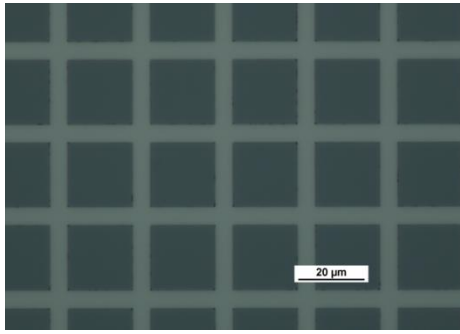
2.9.2(c) (a) 10 times, (b)20 times, (c) 50 times and (d) 100 times of sample 3 after lift-off under optical microscope



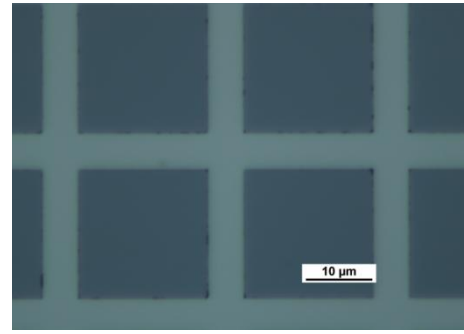
(a)



(b)



(c)



(d)

2.9.2(d) (a) 10 times, (b) 20 times, (c) 50 times and (d) 100 times of sample 4 after lift-off under optical microscope

3 Image Analysis

3.1 Thickness measurement

3.1.1 Basic principles of Atomic Force Microscopy

Atomic Force Microscopy (AFM) is a high-resolution imaging technique used to analyze surface topography at the nanometer and even sub-nanometer scale. Unlike optical or electron microscopy, AFM operates by physically probing the surface with a sharp tip attached to a flexible cantilever. As the tip interacts with the sample, intermolecular forces cause the cantilever to deflect. This deflection is detected via a laser beam reflected into a position-sensitive photodetector. By scanning the tip in a raster pattern and using feedback to maintain constant force, AFM generates a detailed 3D topographic map of the surface. The working principle is shown in Figure 3.1.1. It can operate in air, vacuum, or liquid, making it highly versatile and suitable for various research fields including materials science, nanotechnology, and biology^[19 20].

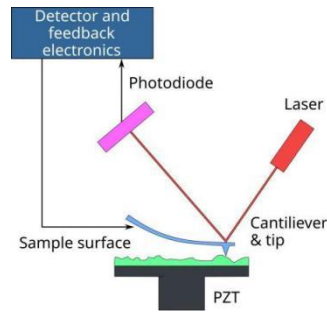


Figure 3.1.1 The working principle of AFM

AFM is widely used in both research and industry for its versatility and nanometer-scale resolution. In materials science, it enables analysis of surface roughness, grain structure, and thin film morphology. In semiconductor fabrication, it helps measure pattern fidelity, etch depth, and line-edge roughness. Its ability to operate in various environments and provide both topographic and mechanical data makes AFM a key tool in nanoscale characterization. As technology advances toward smaller scales, AFM remains essential in nanoscience and nanotechnology research.

3.1.2 Operation process of AFM

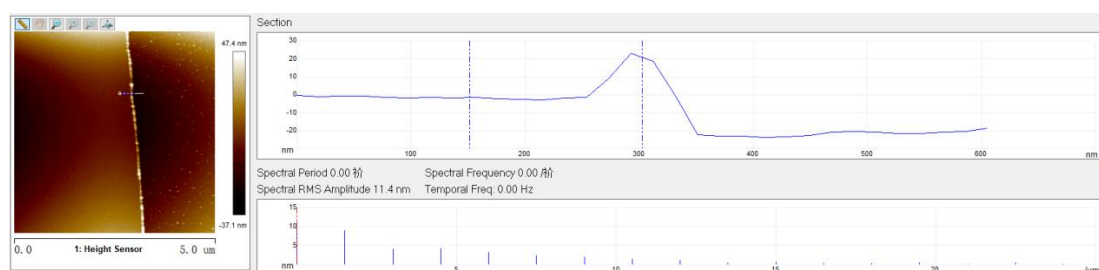
In this experiment, the thicknesses of four different samples were measured using the Bruker Dimension Icon Atomic Force Microscope equipped with ScanAsyst mode. The instrument structure is shown in Figure 3.1.2(a). This instrument provides high-resolution topographic imaging by scanning a sharp probe across the sample surface while detecting deflection variations due to surface features. Each sample was securely mounted on the AFM sample stage, and the scanning was performed under ambient conditions using peak force tapping mode to minimize lateral forces and ensure precise vertical measurements. The resulting height profiles clearly revealed the step edges between the coated and uncoated regions, from which the film thicknesses were extracted. This approach enabled nanometer-scale accuracy

in determining the deposited layer thicknesses on all four samples.

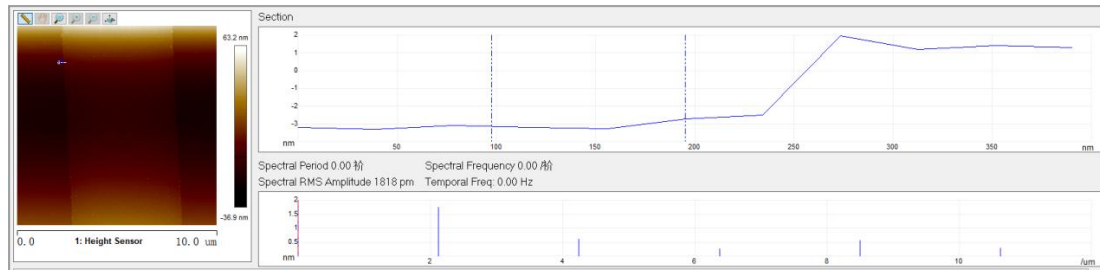


Figure 3.1.2(a) Instrument structure of AFM

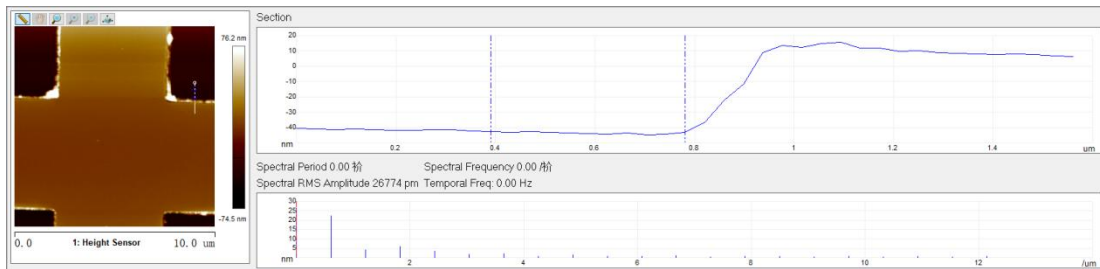
After acquiring AFM images of the samples, NanoScope Analysis 2.0 software was used to analyze and determine the film thickness. By applying the section analysis tool, height profiles were extracted across the boundary between the developed and undeveloped regions of the resist. This allowed for accurate measurement of vertical distances. Based on the results, the thickness of Sample 1 was confirmed to be approximately 20 nm. Sample 2 exhibited a thinner resist layer with a measured thickness of about 5 nm, which is slightly greater than the theoretical value of 2 nm. Samples 3 and 4 both showed thicker coatings, with measured values around 50 nm. These results are consistent with the spin-coating and development conditions used during sample preparation. The measurement results are illustrated in the figures 3.1.2 below.



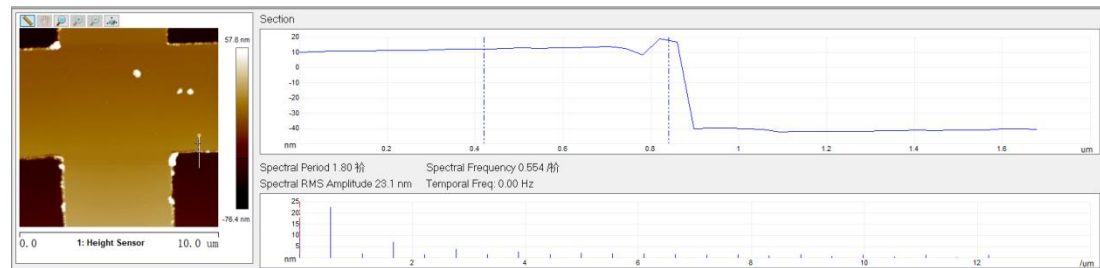
(a)



(b)



(c)



(d)

Figure 3.1.2(b) Software analysis of the thickness of samples measured by AFM

3.2 Sturcture measurement

3.2.1 Basic principles of Scanning Electron Microscope

The Scanning Electron Microscope (SEM) is a highly effective instrument for observing surface morphology with nanometer-scale resolution. It operates by directing a finely focused beam of high-energy electrons onto the surface of a specimen. In standard SEM systems, electrons are emitted from either a thermionic tungsten filament or a field emission source and are accelerated by a high-voltage electric field, typically ranging from 1 to 30 kilovolts. These electrons are then condensed and focused into a narrow probe through a series of electromagnetic lenses. The electron beam is scanned in a raster pattern across the sample

surface, which is maintained under vacuum conditions to prevent scattering by air molecules and ensure beam stability.

When primary electrons interact with atoms in the sample, they generate several types of signals, including secondary electrons, backscattered electrons, and characteristic X-rays. Among these, secondary electrons are the most commonly used for surface imaging due to their low energy, which makes them highly sensitive to surface features. These electrons originate from the top few nanometers of the specimen and are readily influenced by fine changes in topography. Elevated areas and sharp edges emit a greater number of secondary electrons as a result of the edge effect, while flat or recessed areas emit fewer. The emitted electrons are collected by a positively biased detector, which converts the signals into an image where brightness corresponds to the intensity of electron emission, thereby revealing detailed structural features. The basic structure and working principle are illustrated in Figure 3.2.1.

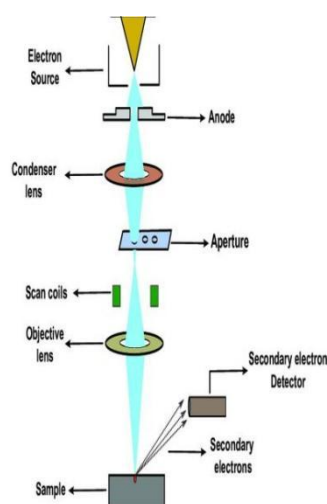


Figure 3.2.1 The basic structure and working principle of SEM

To obtain accurate and high-contrast images, it is essential that the sample surface be electrically conductive. Non-conductive materials tend to accumulate negative charge under

electron irradiation, leading to image distortion or instability in beam alignment. To address this issue, a thin conductive layer of material such as gold, platinum, or carbon is often applied to the sample using a sputter coating process. Furthermore, operation in a high-vacuum environment is critical, as it minimizes unwanted interactions between electrons and gas molecules and helps preserve image resolution and clarity^[21 22].

3.2.2 Basic process of SEM

In this experiment, a JEOL scanning electron microscope was used to image micro- and nanostructures on the sample surface. Before imaging, all samples were inspected for vacuum compatibility. Non-conductive samples were sputter-coated with a thin layer of platinum or gold to prevent surface charging. Clean handling techniques were followed throughout to avoid contamination.

The process began by powering on the system and launching the JEOL software. The stage was moved to the exchange position ($X = 35.0$ mm, $Y = 25.0$ mm, $Z = 8.0$ mm), and the sample was mounted on carbon tape. After sealing the chamber, the “EVAC” button was held to initiate vacuum. Once sufficient vacuum was achieved, the sample was transferred into the chamber using the transfer rod, and proper positioning was confirmed. High voltage (HT) was then activated. When the chamber pressure reached $\sim 9.63\text{E-}5$ Pa, imaging began. The sample was located at low magnification, and the stage was adjusted to center the region of interest. At higher magnification, manual focus and the ACB function were used to optimize image quality. For image capture, “FINE VIEW” mode was selected, followed by “FREEZE” to stabilize the image. Images were saved via the “EXPORT” function. After imaging, the high voltage was turned off and the sample was removed following standard venting procedures.

The following figure 3.2.2 shows the specific operation details of SEM.

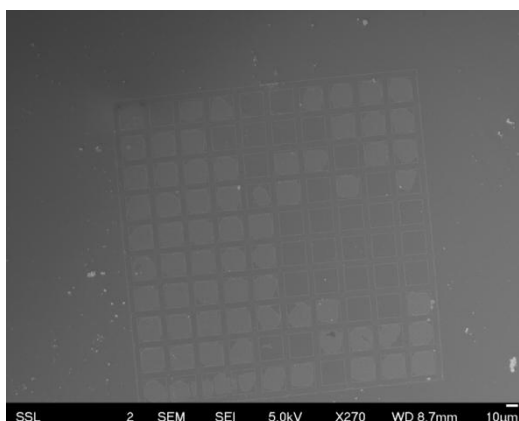


Figure 3.2.2 SEM detailed operation diagram

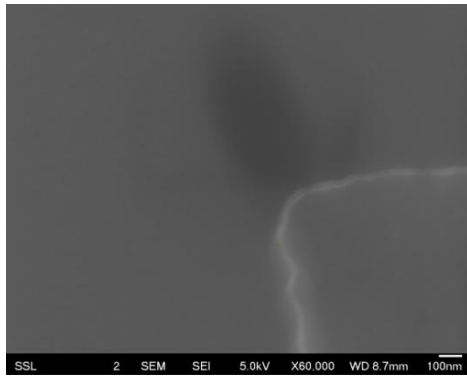
This workflow ensured stable imaging conditions, minimized artifacts, and enabled high-resolution SEM analysis.

3.2.3 Operation process of SEM

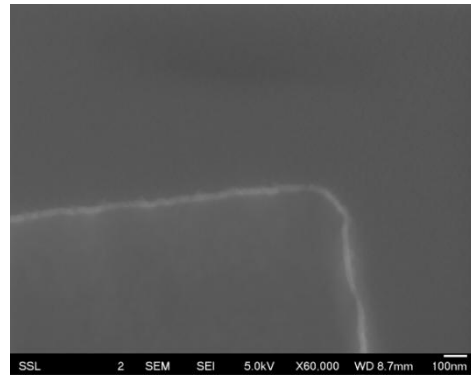
In this experiment, SEM was employed to observe the patterned samples after development and metal deposition. The SEM images clearly reveal the structures and the distinct edges of the patterned regions, allowing for direct assessment of structural integrity and pattern fidelity. Well-defined boundaries and uniform feature formation were observed, indicating successful fabrication. The following images show representative SEM results of different samples acquired under various magnifications.



(a)

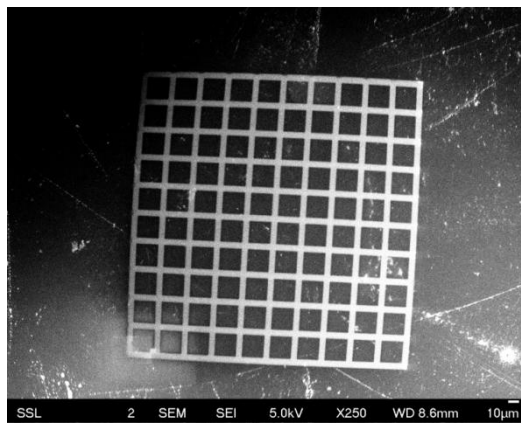


(b)

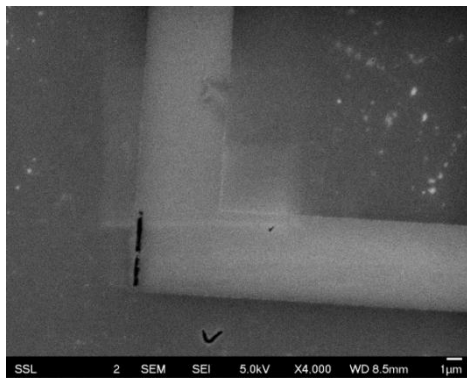


(c)

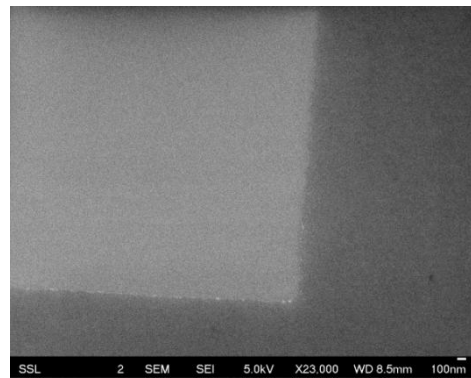
Figure 3.2.3(a) Imaging of Sample 1 under SEM



(a)

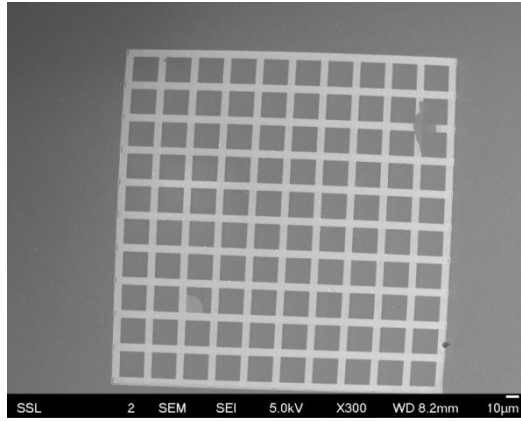


(b)

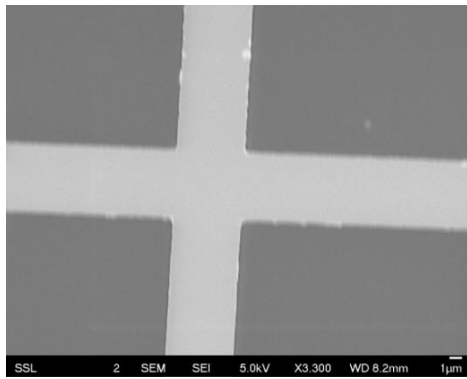


(c)

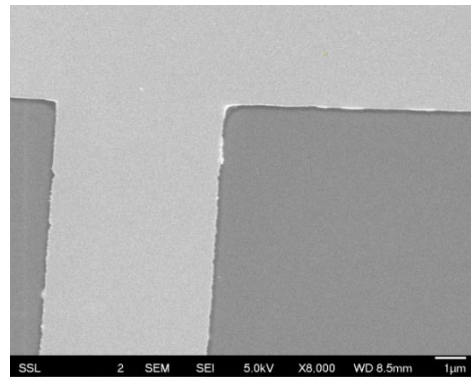
Figure 3.2.3(b) Imaging of Sample 2 under SEM



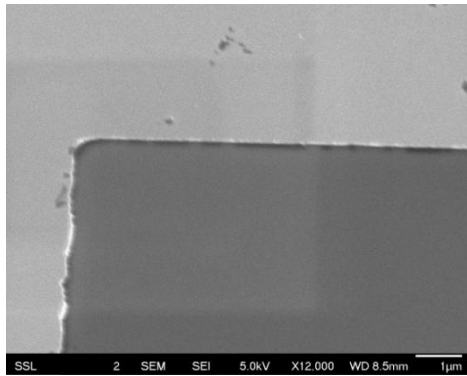
(a)



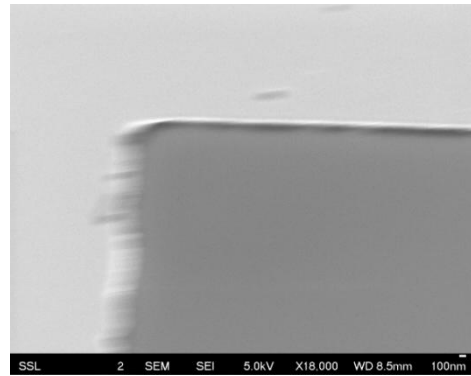
(b)



(c)



(d)



(e)

Figure 3.2.3(c) Imaging of Sample 4 under SEM

3.3 Weakness analysis of samples

3.3.1 Disadvantages of Sputtering platinum

Samples 1 and 2 were prepared by depositing platinum onto PMMA-patterned substrates using the JEOL JFC-1600 auto fine coater. This system employs plasma sputtering, where a

high-voltage argon plasma bombards a platinum target and dislodges atoms that subsequently settle on the sample surface. As a non-directional deposition method, the ejected platinum atoms disperse over a wide range of angles. While this ensures effective coverage of the exposure areas, it also results in unintended deposition on the sidewalls of resist structures.

Such sidewall deposition introduces several issues during development. Firstly, a cap shaped platinum layer may form on the upper edge of the resist, partially sealing features and hindering the developer's ability to reach the lower area. This prevents complete dissolution of the exposed PMMA, leaving behind residual resist, particularly near the edges of grid patterns. These residues can obstruct clean metal lift-off and compromise the dimensional fidelity of the resulting features.

Moreover, obliquely incident atoms often lead to metal overhangs or fringing effects. These appear as irregular fragments or loosely adhered debris along the pattern edges. After development, they remain as metallic artifacts, reducing geometric precision and obscuring the intended pattern boundaries. The combined presence of residual resist and metal fragments leads to blurred edges and diminished SEM image clarity. This blurring hampers both visual interpretation and quantitative analysis, such as evaluating linewidth or assessing pattern uniformity.

In conclusion, although the JEOL JFC-1600 offers a convenient method for depositing conductive films on flat surfaces, its non-directional sputtering process is less suitable for patterned samples with fine features. The resulting issues, including resist residue, metal debris, and poorly defined edges, collectively undermine the performance of Samples 1 and 2 as resolution standards for SEM calibration.

3.3.2 Disadvantages of gold metal layer

During the fabrication of Sample 3 and Sample 4, an additional gold layer was deposited beneath the platinum layer in Sample 3, while Sample 4 did not include this Au layer. Comparative analysis indicates that the presence of exposed Au in Sample 3, particularly in the regions between patterned structures, may negatively impact SEM imaging quality.

The primary concern is that in these unexposure regions, the electron beam directly interacts with the exposed gold surface. This interaction produces secondary electron signals from Au, which differ in yield and energy compared to those from Pt. As a result, the collected image may exhibit reduced contrast, blurred pattern boundaries, and decreased structural clarity due to signal mixing from two distinct metal sources.

Additionally, gold films often possess relatively high surface roughness, especially when deposited under non-optimized sputtering or evaporation conditions. A rough Au surface can lead to uneven nucleation during subsequent Pt deposition, resulting in a non-uniform top metal layer. This irregular morphology may distort the transferred pattern and degrade the overall structural fidelity.

Therefore, considering these factors, sample 4 exhibits better imaging quality and pattern fidelity in SEM observation.

4 Data Analysis

4.1 Analysis method

4.1.1 Core Principle and the Edge-Enhanced SE Emission Phenomenon

During SEM operation, the electron beam is scanned across sharp edges of a standard calibration grid sample. As the beam crosses these edges, the SE signal in both X and Y

directions exhibits distinct transitions, reflecting the spatial distribution of the beam in each axis. These signal profiles are then fitted using mathematical models to extract the FWHM, which quantifies the beam spot size.

The mechanism underlying this signal behavior is known as Enhanced Edge Emission. As illustrated in the figure 4.1.2, when the electron beam irradiates a flat region of the sample, the escape of SEs is limited due to restricted emission angles, and only a portion of the electrons can exit the surface. However, near a sharp structural edge, the geometry provides additional escape paths for SEs. This results in a significant local increase in SE signal, forming strong contrast at the boundaries in the image.

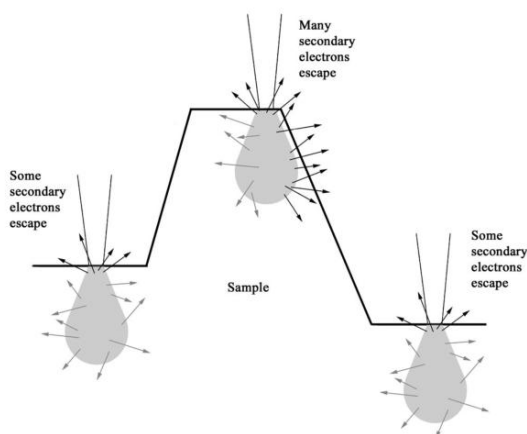


Figure 4.1.1 The Edge-Enhanced SE Emission Phenomenon

4.1.2 Mathematical Model for Beam Spot Size Fitting

To quantitatively determine the spot size, a refined mathematical fitting model was introduced to accurately characterize the variation in secondary electron emission signals as the electron beam scans across sharp structural edges. Traditional models, such as those relying solely on the complementary error function, often fall short when accounting for the “edge enhancement effect” observed in SE data. Therefore, this study adopts a composite model that

incorporates both an error function and a Gaussian peak to more realistically represent the signal profile near edge regions.

The proposed model is expressed as:

$$F = A[1 + \text{Erf}(\frac{2\sqrt{\ln 2}}{f}(d - x))] + B\exp[-\frac{\ln 16}{f^2}(d - x)^2] + C \quad (4.1.2)$$

Where:

x: the scanning position;

f: the Full Width at Half Maximum (FWHM) of the beam spot, the key fitting target;

A: the amplitude of the error function component, simulating the gradual signal increase at the edge;

B: the amplitude of the Gaussian peak, modeling local signal enhancement due to edge-induced SE emission;

C: the baseline signal value representing system background;

d: the physical position of the sharp edge on the calibration grid.

The fitting model used to describe the SE signal during beam scanning includes three distinct components. The first is an error function term, which models the gradual increase in signal intensity as the beam moves across the sample edge. This function effectively captures the cumulative response of a Gaussian-distributed beam profile, representing the smooth transition from unexposed to exposed regions.

The second component is a Gaussian function centered at the edge position d, accounting for the sharp spike in SE emission that occurs at the edge due to enhanced electron escape probability. This localized peak, known as edge-enhanced SE emission, cannot be adequately described by the error function alone, and the Gaussian term improves fitting accuracy around

the edge, refining the FWHM determination.

The third component is a constant baseline that compensates for background SE signal arising from system noise, environmental effects, or emissions from adjacent sample areas. Including this baseline improves the stability and reliability of the fitting process by isolating true beam-induced signal from persistent background contributions.

Overall, these three terms represent global intensity rise, local edge enhancement peak, and constant background, collectively forming an accurate model for SE signal fitting during beam scanning.

In practical application, as the beam scans across a standard test grid, the system records the SE signal intensity at each scanning position x . These intensity data points are then fitted to the above model using nonlinear optimization algorithms. The process yields optimal fitting parameters, especially the FWHM parameter f . Once f is obtained, multiplying it by the actual pixel size yields the physical width of the proton beam spot, allowing for accurate size characterization necessary for precise nanofabrication. The actual pixel size can be obtained by dividing the size of the scanning area by the total number of pixel points.

4.2 Proton beam measurement of sample 4

4.2.1 Basic principles of using proton beam to measure samples

During the process, the high-energy proton beam is precisely focused and scanned through the surface of the sample, where it undergoes inelastic interactions with the atoms of the material. These interactions mainly involve electronic excitation and ionization, resulting in energy transfer that occurs predominantly through electronic stopping as the protons penetrate the near-surface region. This process leads to the emission of various secondary signals, among

which secondary electrons are the most commonly used for image generation.

These low-energy secondary electrons escape from the material surface and are collected by a built-in electron detector, such as the Everhart-Thornley detector. As the scanning system records the position of the beam and the corresponding electron signal at each point, it reconstructs an image pixel by pixel. The brightness of each pixel reflects the detected electron intensity, which directly corresponds to the interaction strength between the beam and the sample at that specific location.

Because the proton beam scans the surface with a controlled step size, the resulting image provides an accurate spatial representation of the beam distribution. This detection method enables precise visualization of beam parameters and is particularly valuable for analyzing beam spot size, resolution at edges, and uniformity of exposure. These capabilities are essential for evaluating and improving the performance of the PBW technique.

4.2.2 Operation process of proton beam measurement

In this experiment, Sample 4 was scanned using the proton beam to obtain spatial information on signal intensity across the surface. The resulting image was constructed based on the number of signals detected at each pixel position, primarily originating from secondary electrons emitted upon proton impact on the platinum surface. These electrons were collected by a detector to generate the two-dimensional grayscale map shown, where pixel brightness indicates the relative signal intensity at that location.

Figure 4.2.2(a) shows the results of scanning the sample 4. The regions with higher brightness in the image indicate stronger interaction between the proton beam and the metal surface, while the darker regions correspond to weaker signals. The cross-shaped pattern observed in

the image reflects either a designed scan path or structural features, and the grayscale gradient along its edges can be further analyzed to extract information about beam spread and resolution.

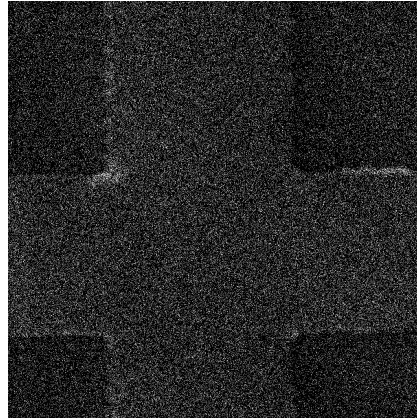


Figure 4.2.2(a) Information obtained from proton beam scanning sample 4

4.2.3 Data analysis of proton beam measurement

In this experiment, focused proton beams were scanned across Sample 4 in both the horizontal (X) and vertical (Y) directions. During the scanning process, the proton beam passed through the sharp edges of a standard test structure, and the resulting secondary electron signals were recorded at each pixel position. The signal intensity corresponding to each pixel reflects the number of SEs emitted from that specific location, which is primarily influenced by the local beam intensity and the geometric features near the edge. Figure 4.2.3(a) shows the specific operation of scanning the position of sample 4.

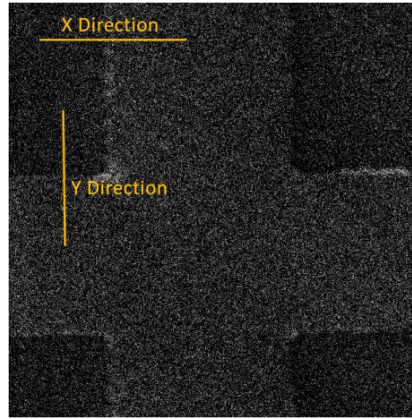


Figure 4.2.3(a) Position of Sample 4 scanned by PBW

The collected raw data consists of the pixel positions and their corresponding SE signal intensities. These data were processed using Origin software. To accurately capture the gradual intensity transition and localized signal enhancement near the edge regions, the experimental data were fitted using the previously defined fitting model (Equation 4.1.3). The most important fitting parameter is f , which represents the full width at half maximum of the beam profile. By multiplying the FWHM (in pixel units) by the physical pixel width, the actual beam spot size in physical dimensions can be calculated.

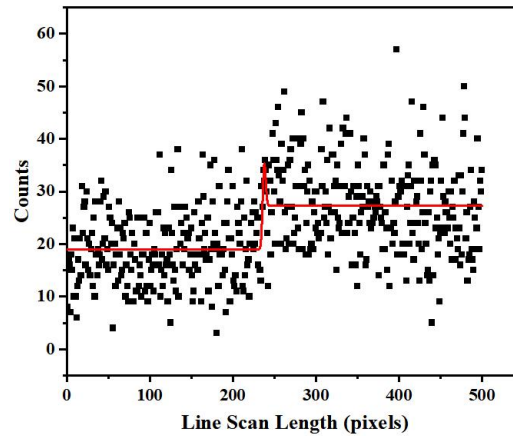


Figure 4.2.3(b) Relationship diagram between pixels in the horizontal scanning area and the received
signal intensity

Table 4.2.3(a) The parameter results in the equation fitted using Origin software based on the data

Model	Fitting Function
Equation	$F = A[1 + \text{Erf}(\frac{2\sqrt{\ln 2}}{f}(d - x))] + B\exp[-\frac{\ln 16}{f^2}(d - x)^2] + C$
A	-4.1 ± 0.3
B	12 ± 6
C	27.2 ± 0.4
d	237 ± 1
f	4.23 ± 2
R ²	0.234
Adjusted R ²	0.228

Figure 4.2.3 (b) shows the data of proton beam scanning the edge of sample 4 in the horizontal direction. The red curve is the result fitted according to equation 4.1.3. The specific fitting parameters are shown in Table 4.2.3(a). The value of f representing FWHM is 4.23 ± 2 pixels. For the X direction, the total physical scan length was $14.3474 \mu\text{m}$, with 1024 pixel points. Thus, the actual value of FWHM is $60 \pm 20 \text{ nm}$, the calculated beam spot size in the horizontal direction is $30 \pm 10 \text{ nm}$.

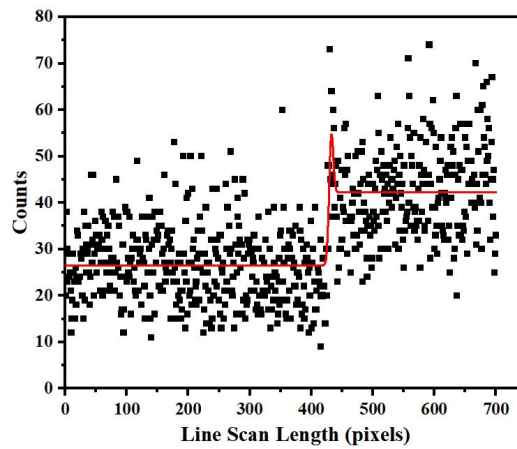


Figure 4.2.3(c) Relationship diagram between pixels in the vertical scanning area and the received signal intensity

Table 4.2.3(b) The parameter results in the equation fitted using Origin software based on the data

Model	Fitting Function
Equation	$F = A[1 + \text{Erf}(\frac{2\sqrt{\ln 2}}{f}(d - x))] + \text{Bexp}[-\frac{\ln 16}{f^2}(d - x)^2] + C$
A	-7.8 ± 0.3
B	19 ± 5
C	42.2 ± 0.5
d	431 ± 1
f	7.93 ± 2
R ²	0.439
Adjusted R ²	0.435

Figure 4.2.3 (c) shows the data of proton beam scanning the edge of sample 4 in the horizontal direction. The red curve is the result fitted according to equation 5.1.3. The specific fitting parameters are shown in Table 4.2.3(b). The value of f representing FWHM is 7.93 ± 2 pixels. For the Y direction, the scan covered a physical length of $13.0189 \mu\text{m}$, also with 1024 pixel points. Thus, the actual value of FWHM is $100 \pm 20 \text{ nm}$, the calculated beam spot size in the vertical direction is $50 \pm 10 \text{ nm}$.

These extracted values provide a quantitative evaluation of the spatial resolution of proton beams in two scanning directions and provide important references for evaluating the performance of beam focusing systems used in proton beam writing.

4.3 Error analysis

There may be some errors during proton beam scanning of samples. The main reason for the error is that the data obtained from Proton beams scanning is the number of electrons detected at each pixel, so some pixels may display a quantity of 0. Therefore, when processing the data, it is necessary to sum up multiple sets of data before calculating. Besides, during the scanning

process, the sample was not placed completely vertically. Therefore, adding multiple rows of data for calculation will result in an increase in edge width, leading to a larger calculated beam spot size.

But this issue does not need to be considered when measuring the SEM beam spot size. Because using SEM to measure the sample will result in a scanned image. When processing data, image information need to be converted into grayscale values. Grayscale value can be used to describe the brightness intensity of each pixel in an image. At this point, each pixel has corresponding grayscale value data, so only one set of data can be analyzed without the need to add multiple rows of data for calculation, it can improve the accuracy of the calculation results.

4.4 SEM measurement of sample 4

When measuring SEM data, the value of parameter f can be obtained by selecting a set of data in the x and y directions and converting them into grayscale values, and fitting them using a formula. Then, based on the actual size represented by each pixel, calculate the beam spot size in the x and y directions.

During the scanning process, if the SEM is not properly focused or if parameters such as working distance and astigmatism correction are not optimally set, it may result in the appearance of shadows along the edges of structures in the image. This phenomenon commonly referred to as the “shadow effect”. This effect degrades image quality by creating artificial contrast or edge distortions, which can interfere with accurate interpretation and quantitative analysis. In particular, when performing edge-based data fitting to extract parameters such as beam spot size, the presence of shadow artifacts can introduce fitting

errors or skew the results. Therefore, in this work, image regions affected by shadowing were carefully identified and excluded from the fitting process to ensure the reliability of the extracted data.

4.4.1 Measurement of SEM in E6

Firstly, I scanned the sample using SEM (model: Tescan GAIA3 Triglav) and obtained an image, as shown in Figure 4.4.1(a). According to the information in the figure, the scale bar is $2\mu\text{m}$, corresponding to 369 pixel points, so the actual size of each pixel point is about 5nm. Afterwards, ten sets of data were selected along the x and y directions, and the size of the beam spot was fitted. The specific results are shown in Table 4.4.1(a) and 4.4.1(b).

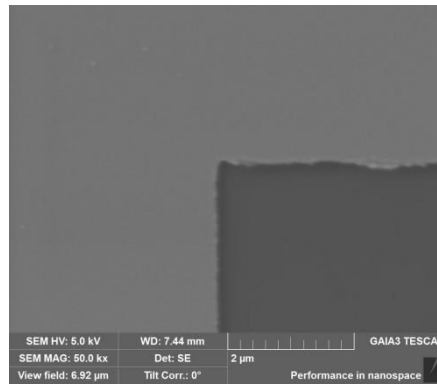


Figure 4.4.1(a) Sample structure diagram observed under SEM

Table 4.4.1(a) Calculation results of horizontal beam spot size

	1	2	3	4	5	6	7	8	9	10
f/(pixel)	10.8	8.4	9	8.4	11.3	8.4	8.8	9.1	8.6	8.8
pixel size/(nm)	5									
Beam spot size/(nm)	27	21	22.5	21	28.25	21	22	22.75	21.5	22

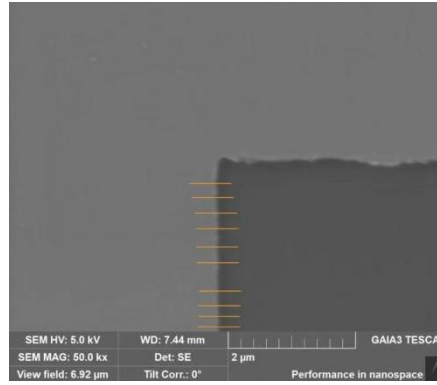


Figure 4.4.1(b) Select the position of data in the horizontal direction

Table 4.4.1(b) Calculation results of vertical beam spot size

	1	2	3	4	5	6	7	8	9	10
f/(pixel)	10.1	9.5	9.7	8.7	11.5	9.2	9.6	11.7	11.5	9.8
pixel size/(nm)	5									
Beam spot size/(nm)	25.25	23.75	24.25	21.75	28.75	23	24	29.25	28.75	24.5

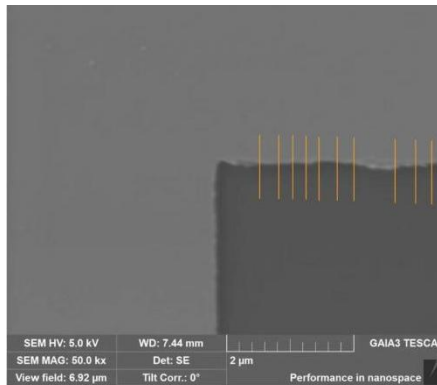
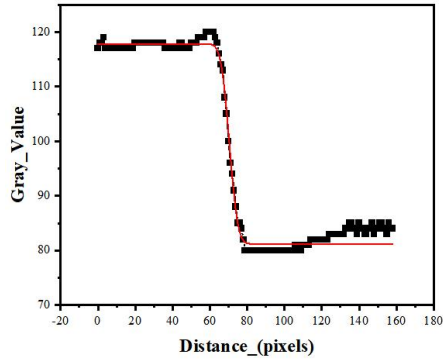


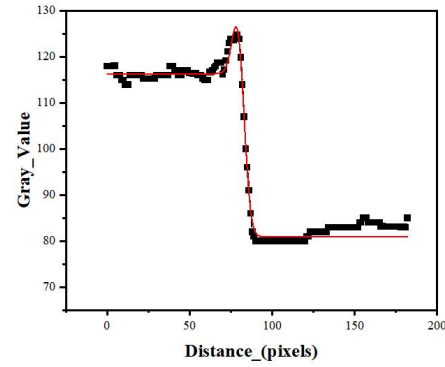
Figure 4.4.1(c) Select the position of data in the vertical direction

Based on the above data, the minimum beam spot size that can be calculated in the x direction is 21nm, and the minimum beam spot size that can be calculated in the y direction is 21.75nm.

The fitted data image is shown in Figure 4.4.1(d).



(a)



(b)

Figure 4.4.1(d) The best data fitting image in (a) horizontal direction and (b) vertical direction

Moreover, the SEM system shows a beam spot size of 21nm. According to the measurement results, the best data calculated in the x and y directions is also around 21nm. This proves that the calculation accuracy of this sample is very high.

4.4.2 Measurement of SEM in Physics Department

I also used SEM (model: JEOL JSM-6700F) to scan the sample and obtained an image, as shown in Figure 4.4.2(a). According to the information in the figure, the scale bar is 1 μ m, corresponding to a pixel count of 250, so the actual size of each pixel is about 4nm. Afterwards, ten sets of data were selected along the x and y directions, and the size of the beam spot was fitted. The specific results are shown in Table 4.4.2(a) and 4.4.2(b).

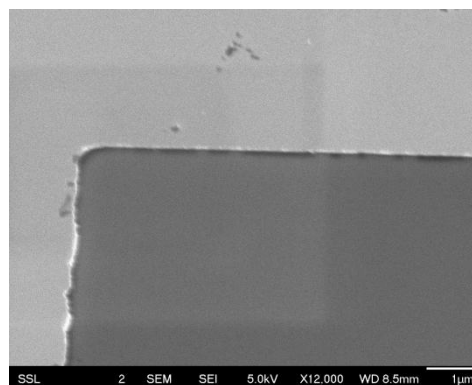


Figure 4.4.2(a) Sample structure diagram observed under SEM

Table 4.4.2(a) Calculation results of horizontal beam spot size

	1	2	3	4	5	6	7	8	9	10
f/(pixel)	7.5	7	10.2	8.8	10.3	8.4	8.1	10.2	6.9	8.8
pixel size/(nm)	4									
Beam spot size/(nm)	15	14	20.4	17.6	20.6	16.8	16.2	20.4	13.8	17.6

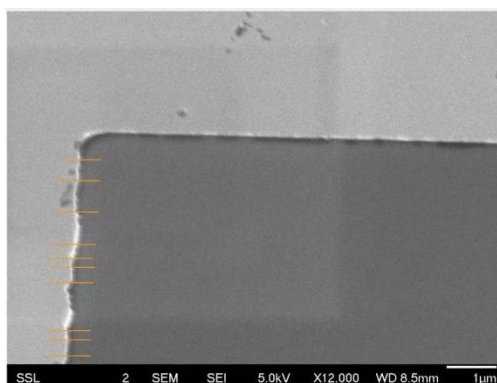


Figure 4.4.2(b) Select the position of data in the horizontal direction

Table 4.4.2(b) Calculation results of vertical beam spot size

	1	2	3	4	5	6	7	8	9	10
f/(pixel)	11.4	11.8	9.2	10	12.5	9	13.2	17.1	17.3	14
pixel size/(nm)	4									
Beam spot size/(nm)	22.8	23.6	18.4	20	25	18	23.6	34.2	34.6	28

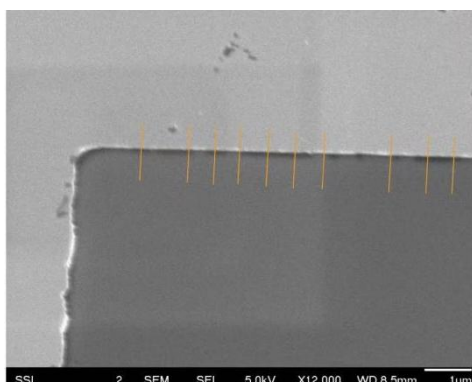


Figure 4.4.2(c) Select the position of data in the vertical direction

Based on the above data, the minimum beam spot size that can be calculated in the x direction is 13.8nm, and the minimum beam spot size that can be calculated in the y direction is 18nm.

The fitted data image is shown in Figure 4.4.2(d).

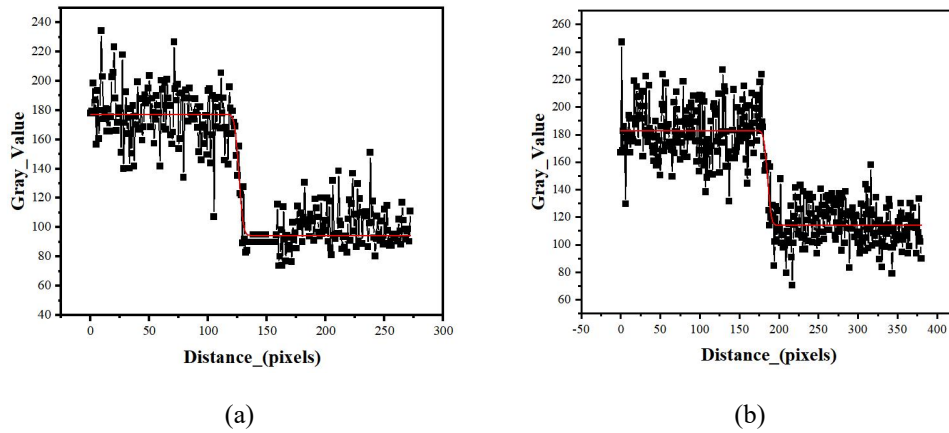


Figure 4.4.2(d) The best data fitting image in (a) horizontal direction and (b) vertical direction

4.5 Sample as a Beam Focus Reference

In addition, this sample has another function, it can help confirm whether the electron beam is well-focused during imaging.

As we can see in the figure 4.5a, the vertical edge of the metal structure appears very sharp and clearly defined. When the SEM beam is properly focused, the boundary between the metal and the background becomes crisp, and the edge profile looks steep. This indicates that in the y-direction, the beam is tightly focused, producing high-resolution images with good edge clarity. In contrast, the horizontal edge appears slightly more blurred, and the edge transition is wider. This suggests that the beam may not be as well-focused in the x-direction, possible due to alignment issues, stage tilt, or uncorrected astigmatism.

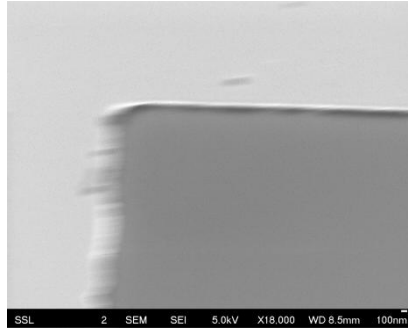
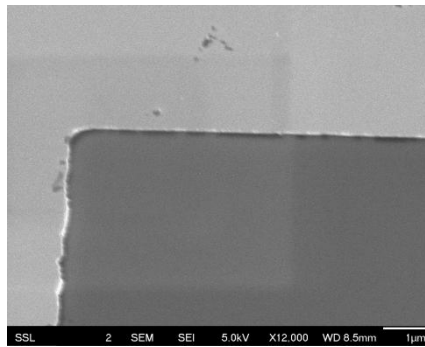


Figure 4.5(a) Imaging when the beam spot is not adjusted to the optimal state

By comparing the edge sharpness along the x and y directions, we can evaluate the beam focus. If both directions show sharp, clean edges, like figure b, we can be confident that the SEM is in optimal focus. But if one direction is less defined, it may indicate the need to adjust working distance, fine-tune the focus and so on.



4.5(b) Imaging when the beam spot is adjusted to the optimal state

5 Conclusion

This project successfully developed a grid-like resolution standard sample for SEM calibration. Using lithography with PMMA and platinum deposition, well-defined micro- and nanoscale patterns were fabricated with high edge fidelity. Key parameters such as spin-coating, exposure dose, and development time were optimized to ensure accurate pattern transfer.

The fabricated structures were evaluated using optical microscopy, AFM, and SEM, confirming their dimensional integrity. Additionally, In addition, This sample enables us to

measure the SEM beam spot size with high accuracy. And it also serves as a practical reference to help verify whether the beam is in optimal focus during operation.

In summary, the project presents a reproducible method for producing high-quality SEM resolution standards. The combination of precise fabrication and quantitative validation makes these samples valuable tools for SEM calibration in nanotechnology and materials research.

6 References

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